

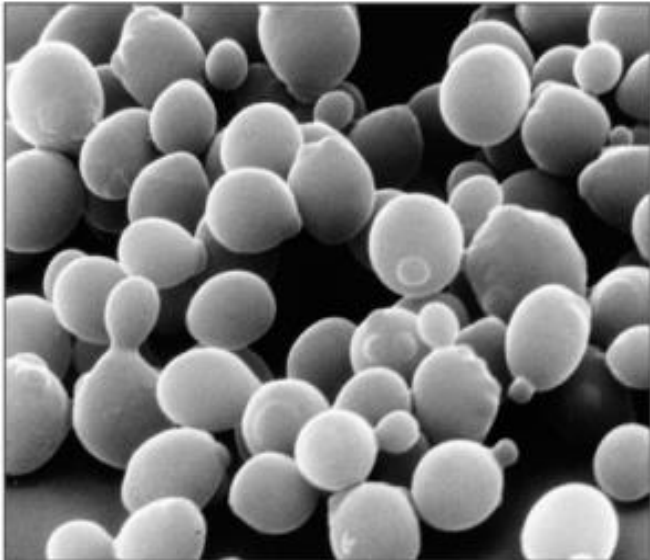
Dynamical Systems Modeling and Visualization of Yeast Cold Shock Gene Regulatory Networks

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UCI
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Cold shock is an environmental stress that is not well-studied

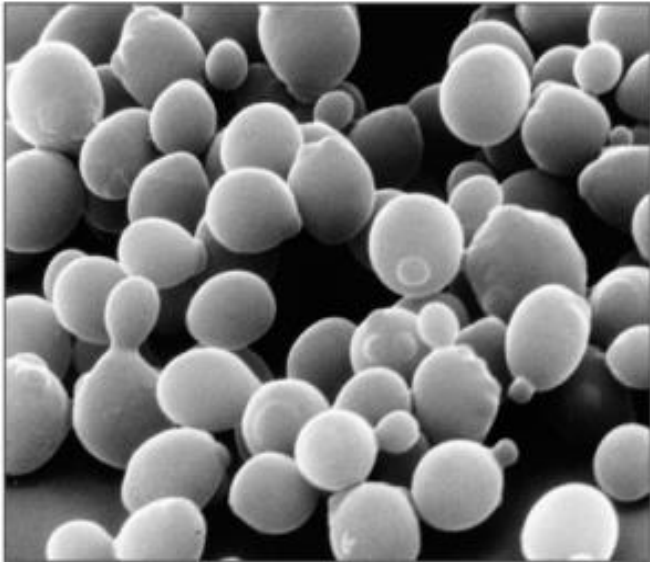


Alberts *et al.* (2004)

10 μm

- Cold shock temperature range for yeast is 10-18°C.
- The cold shock response can be divided into an early and late response.
 - General Environmental Stress Response (ESR) genes are induced in the late response.
 - Late response is regulated by the Msn2/Msn4 transcription factors.
 - No “canonical” transcription factor responsible for the early response.

Yeast respond to cold shock by changing gene expression



Alberts *et al.* (2004)

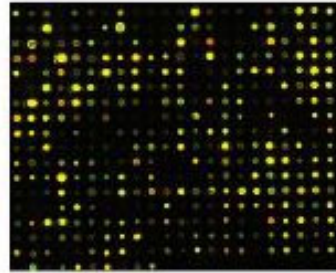
10 μm

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 - Late response is regulated by the Msn2/Msn4 transcription factors.
 - No “canonical” transcription factor responsible for the early response.

1. Which transcription factors control the early response?
2. What are their relative levels of influence?
3. What are the indirect effects of other transcription factors in the *gene regulatory network*?

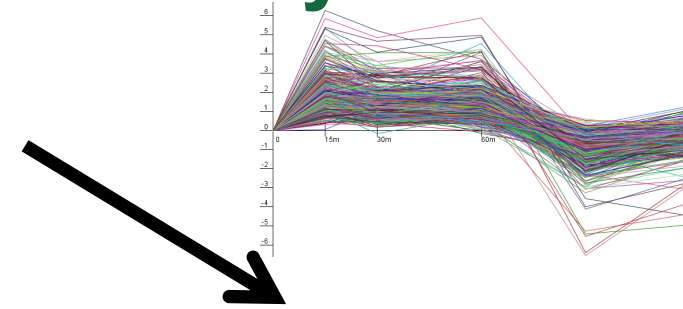
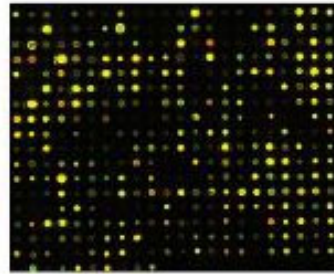
Systems biology approach to understanding the regulation of the cold shock response in yeast

Cold shock microarray
data from wt and TF
deletion strains



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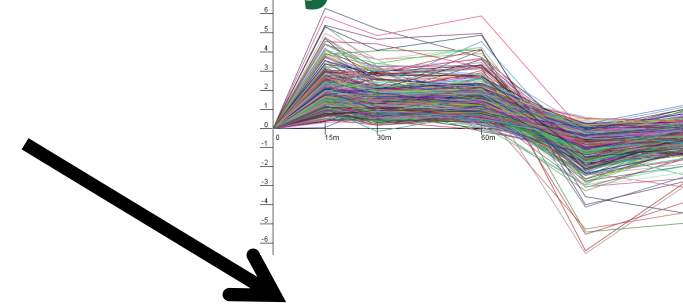
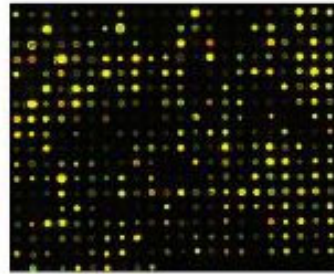
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Normalization,
statistical analysis,
clustering

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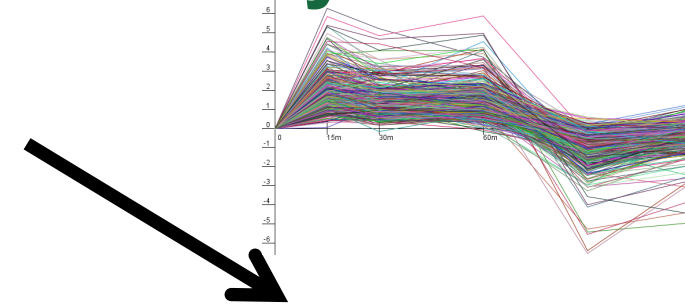
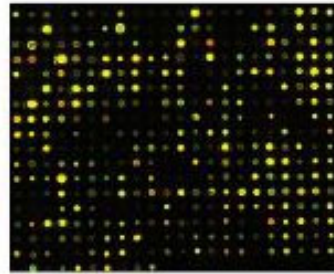


Normalization,
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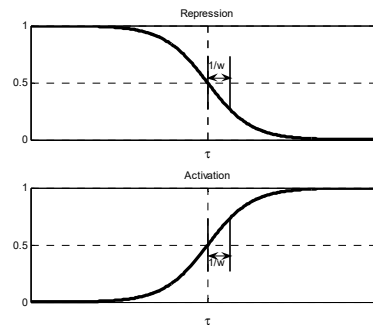
Derivation of gene
regulatory networks
from YEASTRACT

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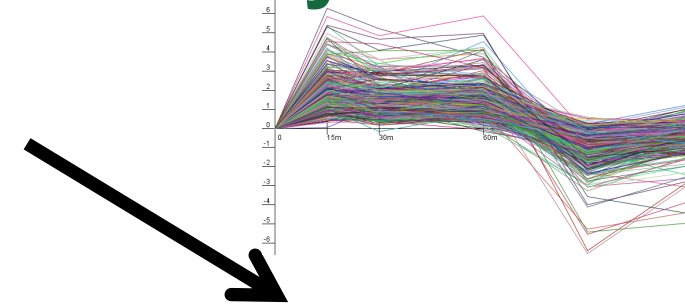
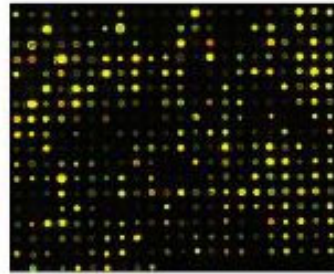


Dynamical systems
modeling using
GRNmap

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Systems biology approach to understanding the regulation of the cold shock response in yeast

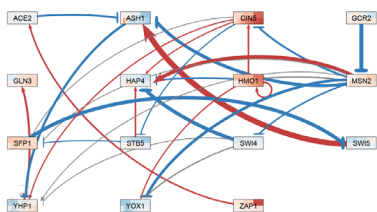
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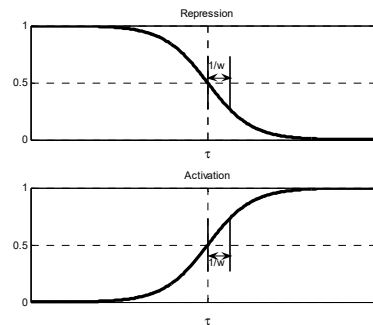
Normalization, statistical analysis, clustering

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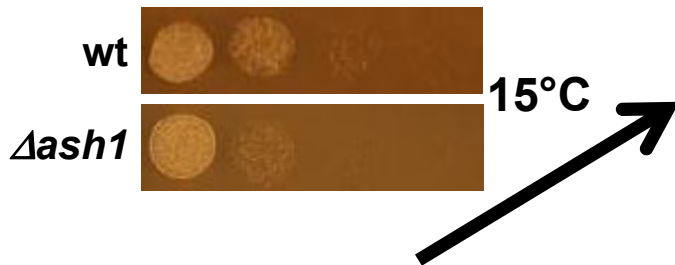
Visualization of modeling results using GRNsight



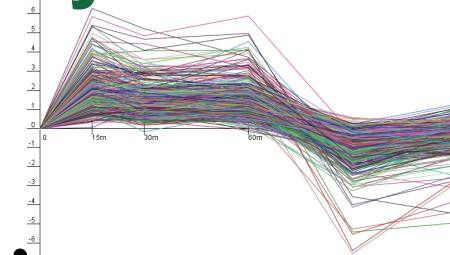
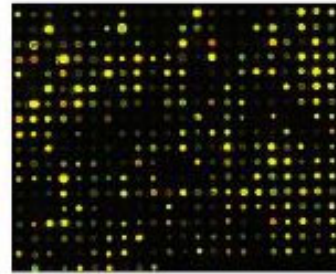
Dynamical systems modeling using GRNmap



Systems biology approach to understanding the regulation of the cold shock response in yeast



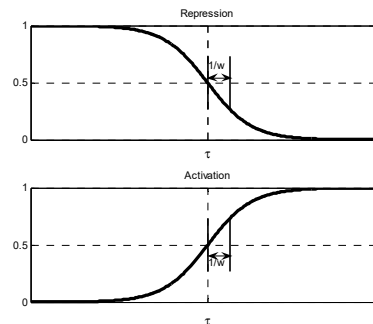
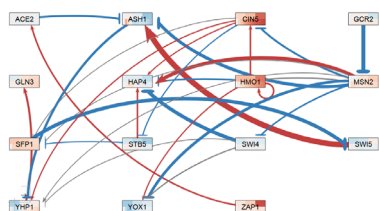
Cold shock microarray data from wt and TF deletion strains



Interpretation,
new questions,
new experiments

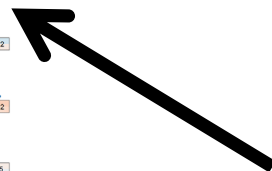
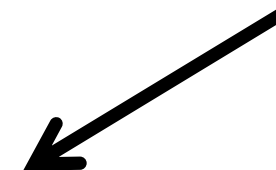
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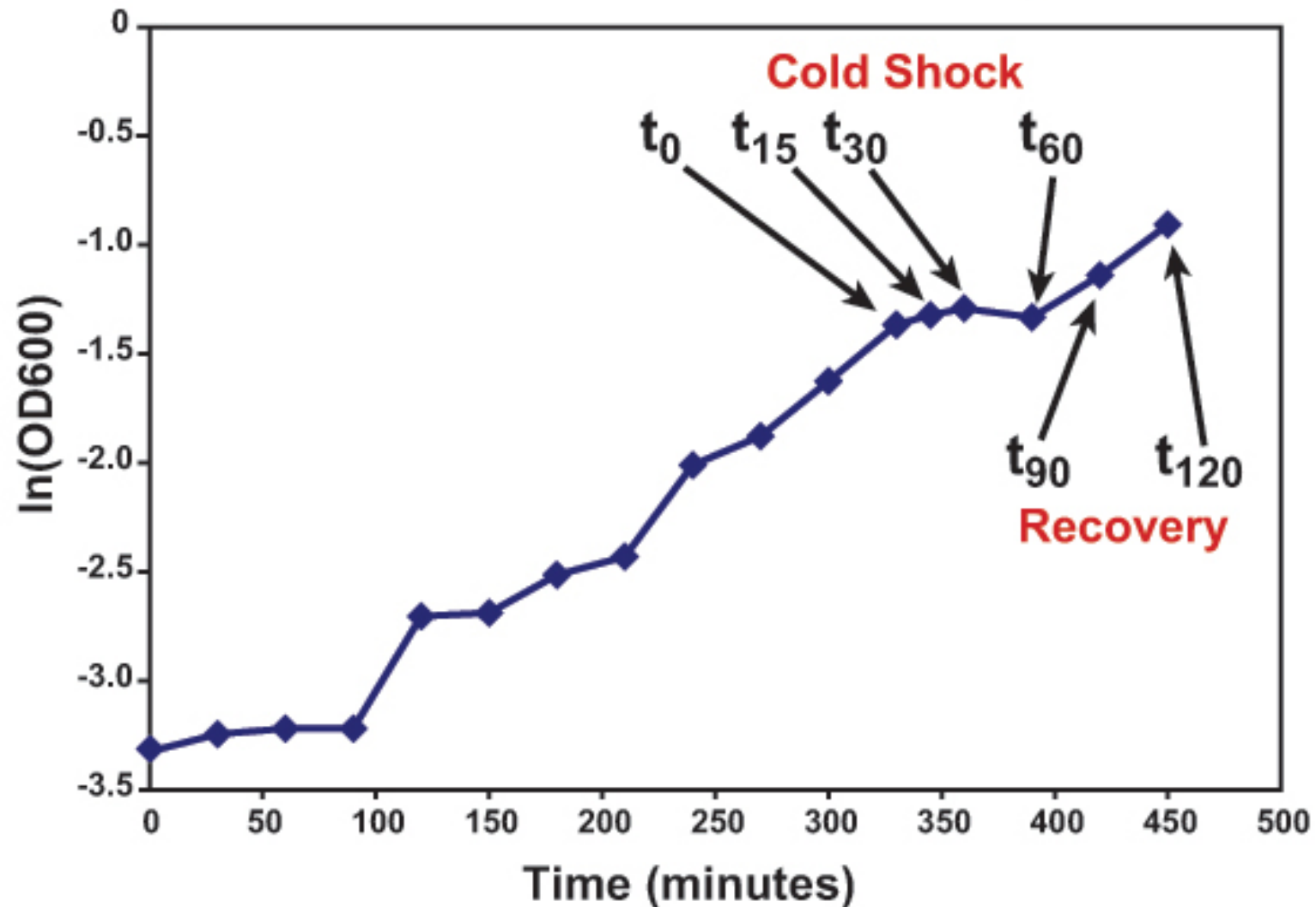


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Derivation of gene
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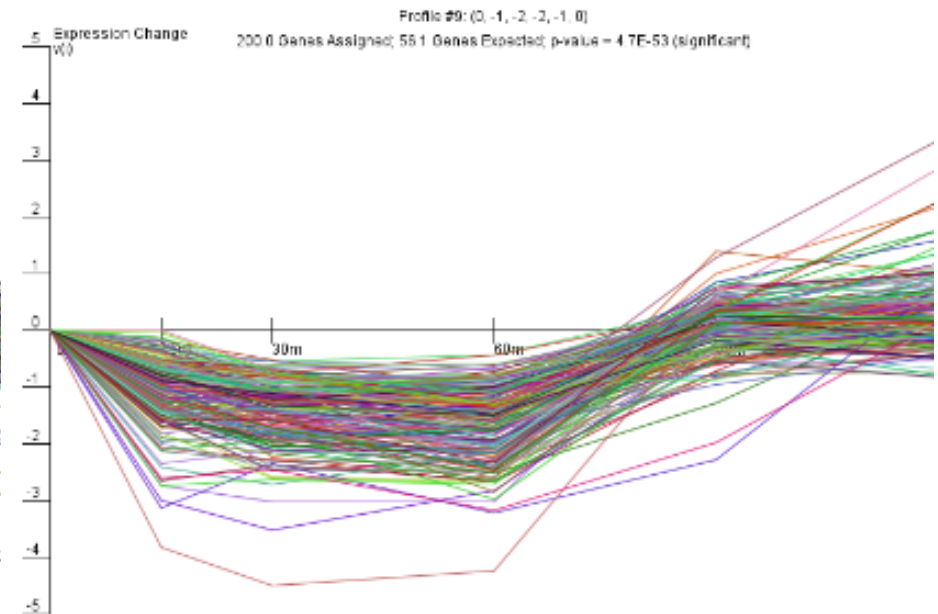
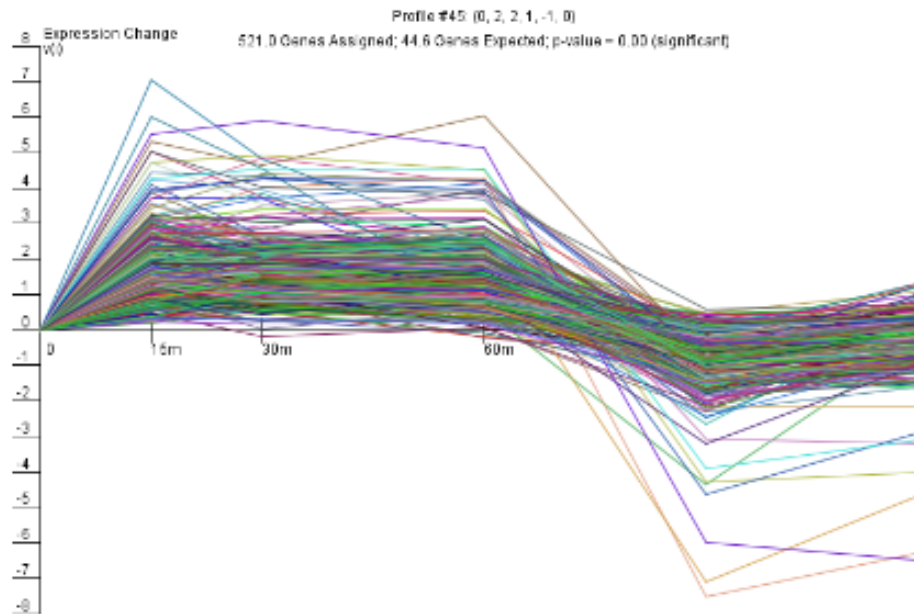


Yeast cells were harvested for microarrays before, during, and after a cold shock and during recovery



Two dominant expression profiles respond to cold shock and recovery

Expression ($\log_2$ fold change)



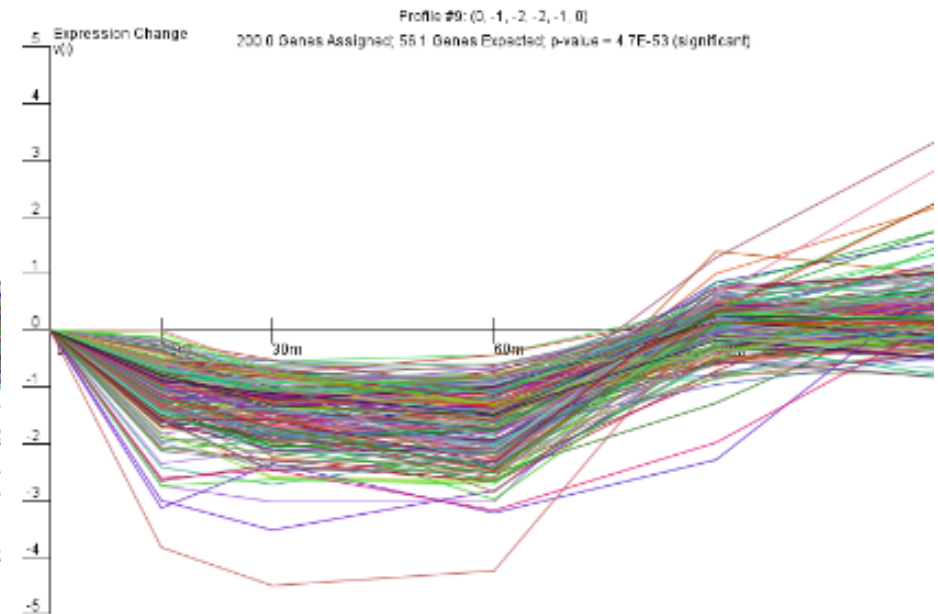
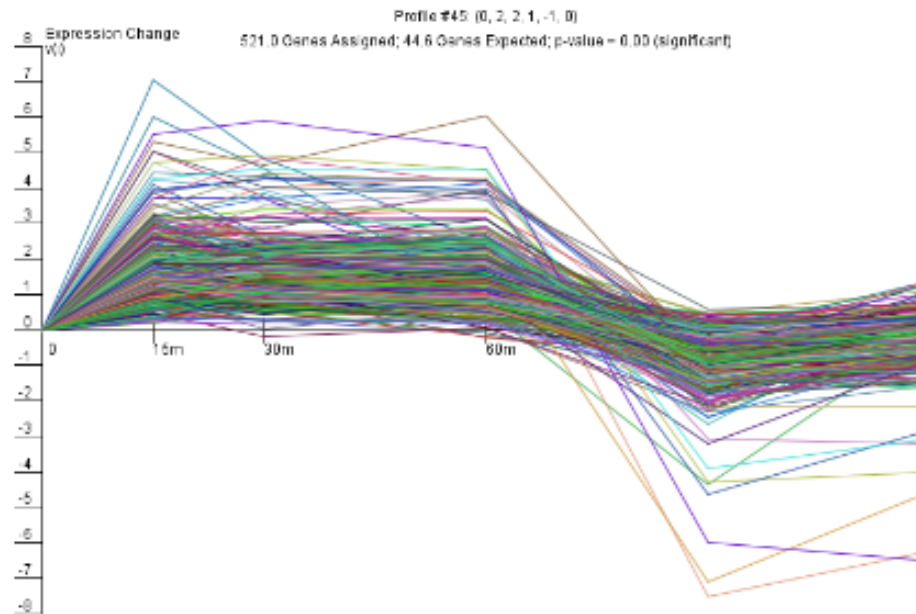
Time (minutes)

Gene Ontology categories assigned to clusters:

- Ribosome biogenesis
- Zinc ion homeostasis
- Hexose transport
- Endomembrane system
- Protein and vesicle transport
- Negative regulation of nitrogen compound process

Genes with similar expression profiles are regulated by the same transcription factor

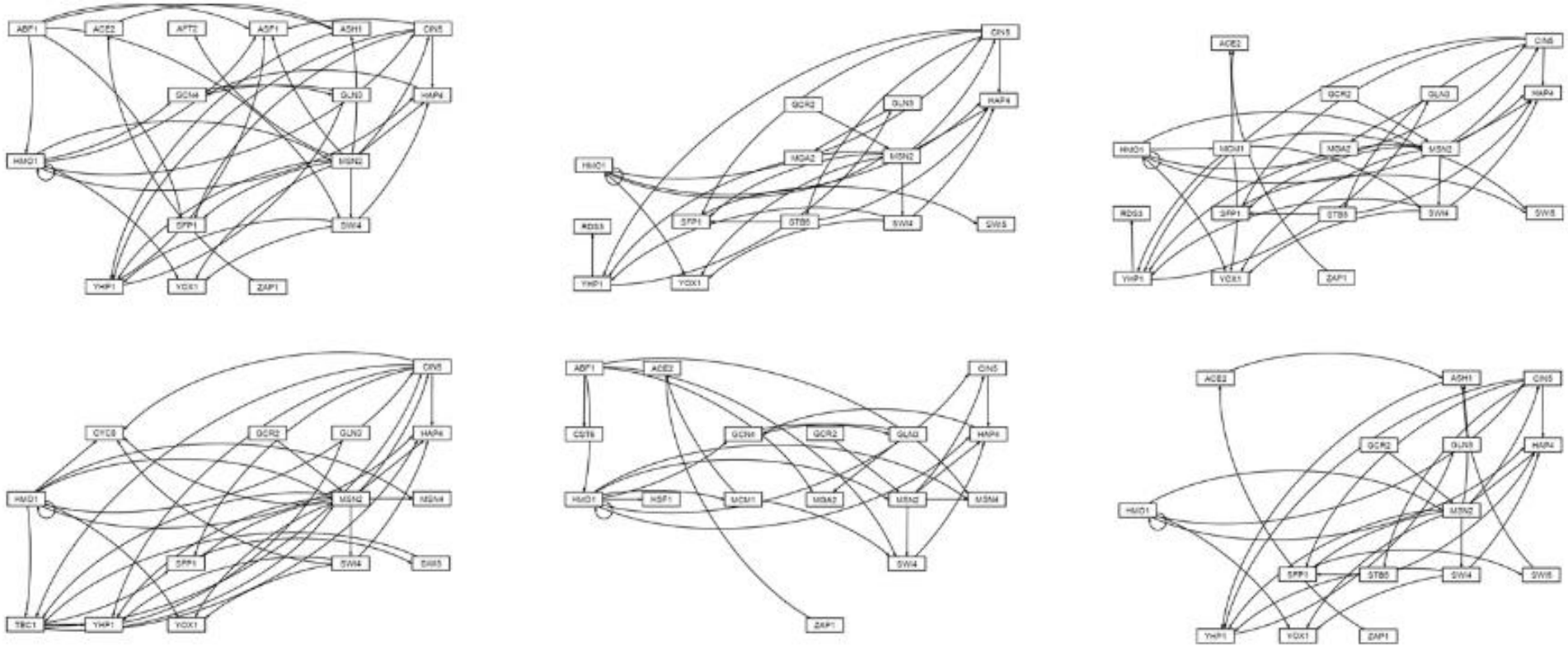
Expression ($\log_2$ fold change)



Time (minutes)

Which transcription factors regulate these genes?

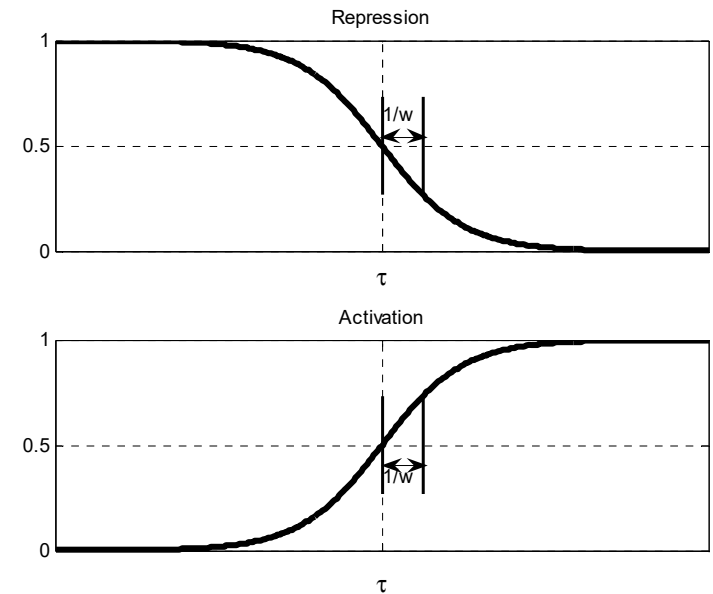
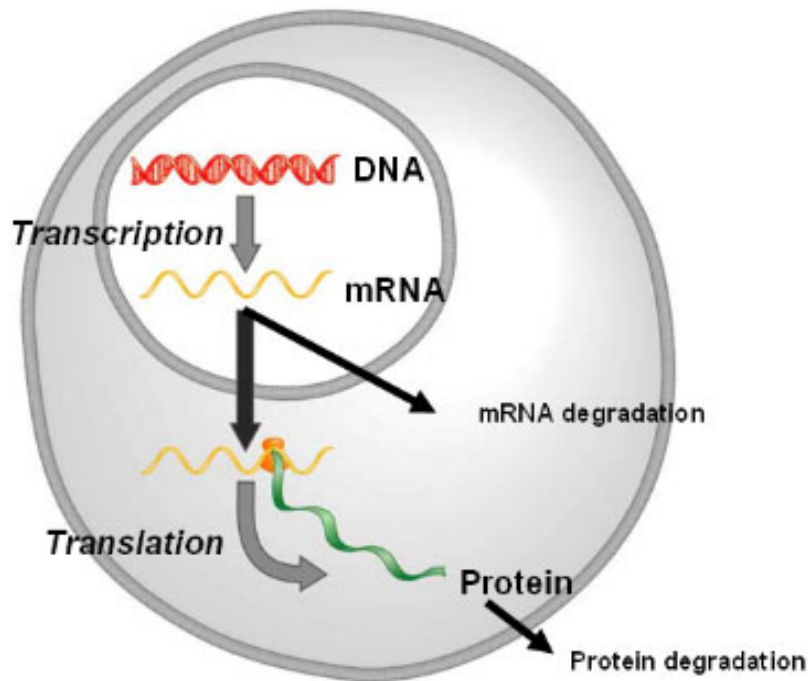
Six candidate GRNs were derived from the YEASTRACT database



- Clusters of genes from each strain's stem analysis were input into YEASTRACT to find potential regulators.
- The top 15 potential regulators for each strain's data were used to generate a regulation matrix.
- Transcription factors for which we had deletion strain microarray data were also included.

GRNmap: Gene Regulatory Network Modeling And Parameter estimation

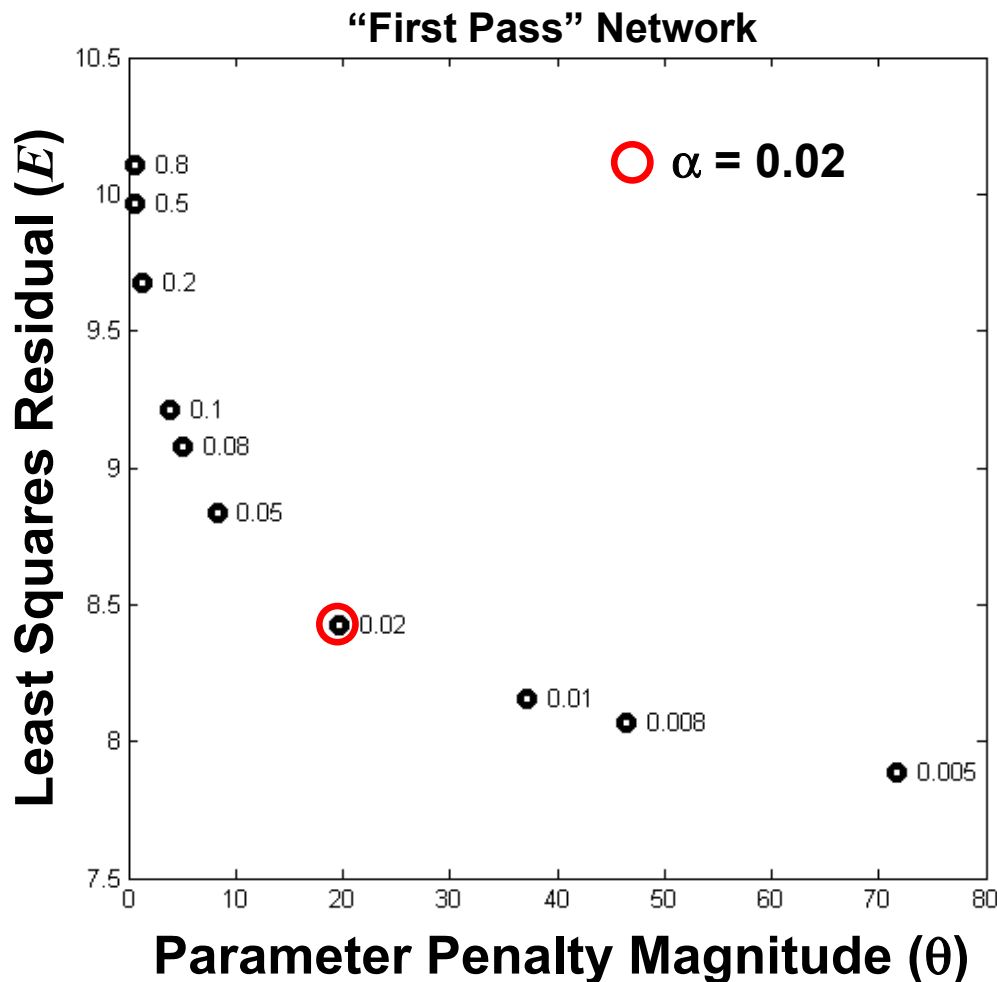
$$\frac{dx_i(t)}{dt} = \frac{P_i}{1 + \exp\left(-\left(\sum_j (w_{ij}x_j(t)) - b_i\right)\right)} - d_i x_i(t)$$



- The change in expression of one gene is *production – degradation*.
- Weight parameter, w , gives the direction (activation or repression) and magnitude of regulatory relationship.

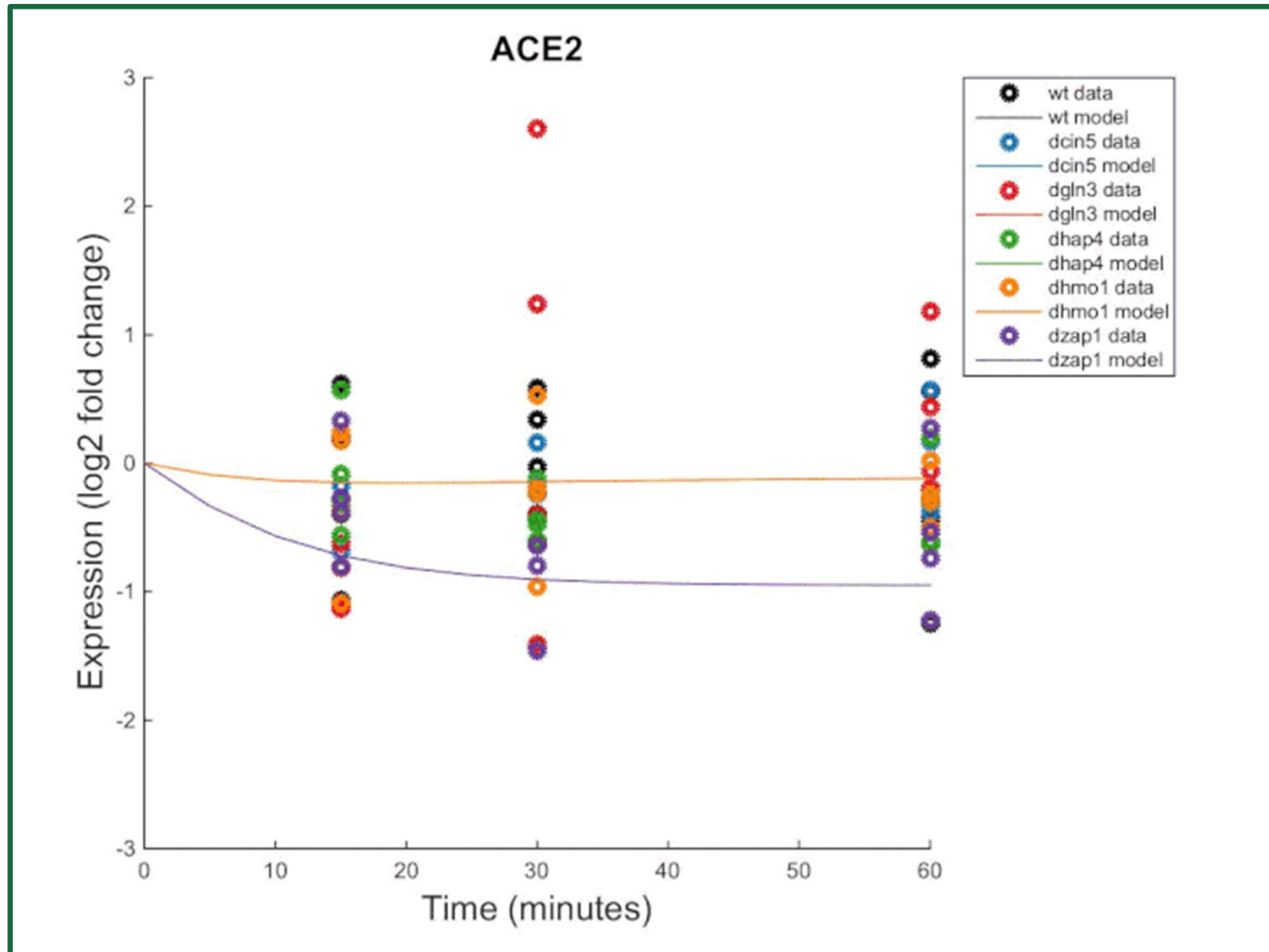
Parameter values are estimated from the time course expression data using a least squares error function

$$E = \frac{1}{n_g n_t} \sum_{i=1}^{n_g} \sum_{k=1}^{n_t} [z^d(t_r) - z^c(t_r)]^2 + \alpha \|\theta\|^2$$

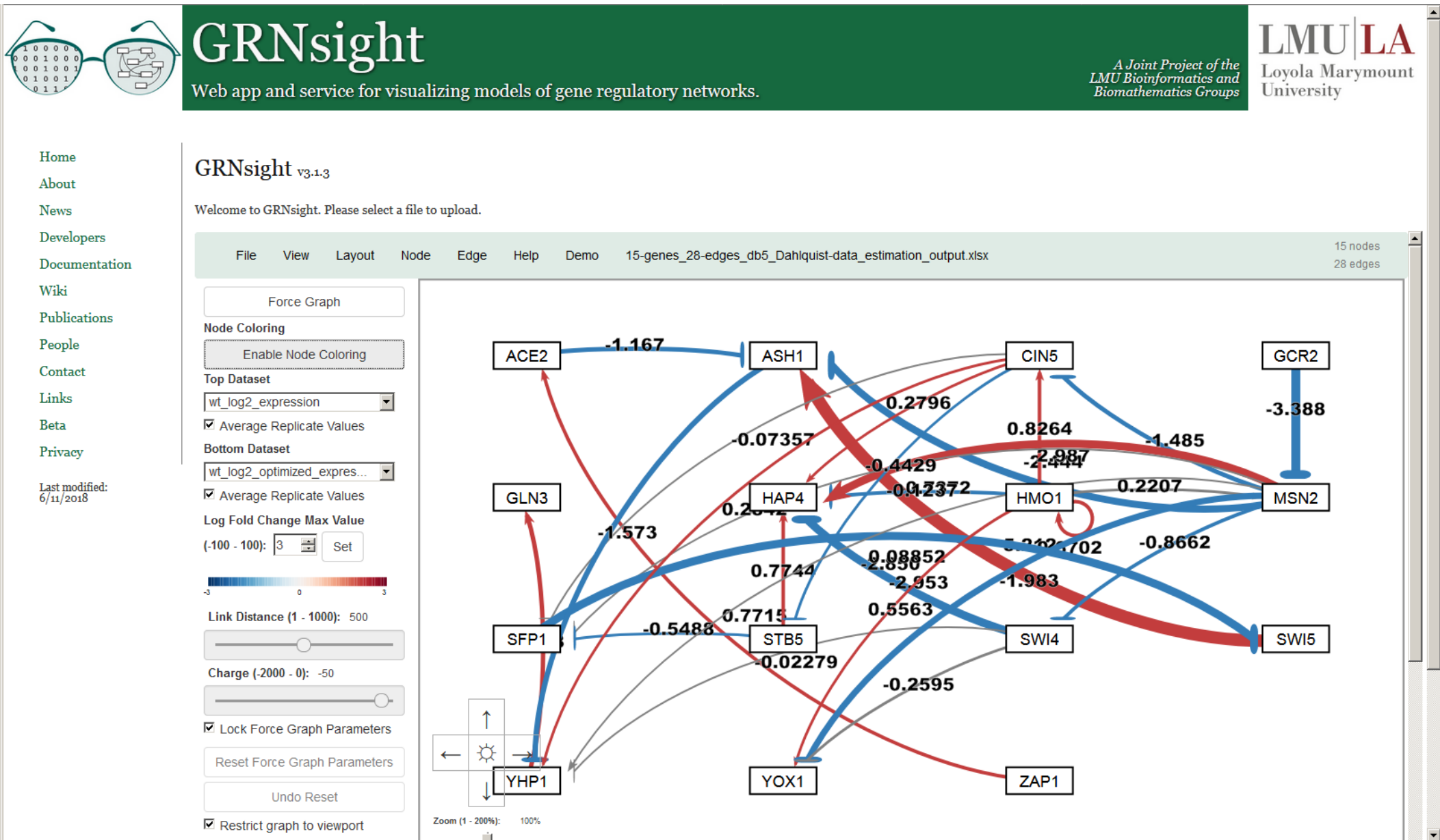


- Total number of parameters is (2 X no. of genes) + no. of edges.
- A penalty term forces MATLAB's optimization algorithm to minimize the function.
- θ is the combined production rate, weight, and threshold parameters.
- α is determined from the “elbow” of the L-curve.
- E represents the difference between the microarray data and simulated data generated by the model.

Forward simulation of the model fits the expression data well

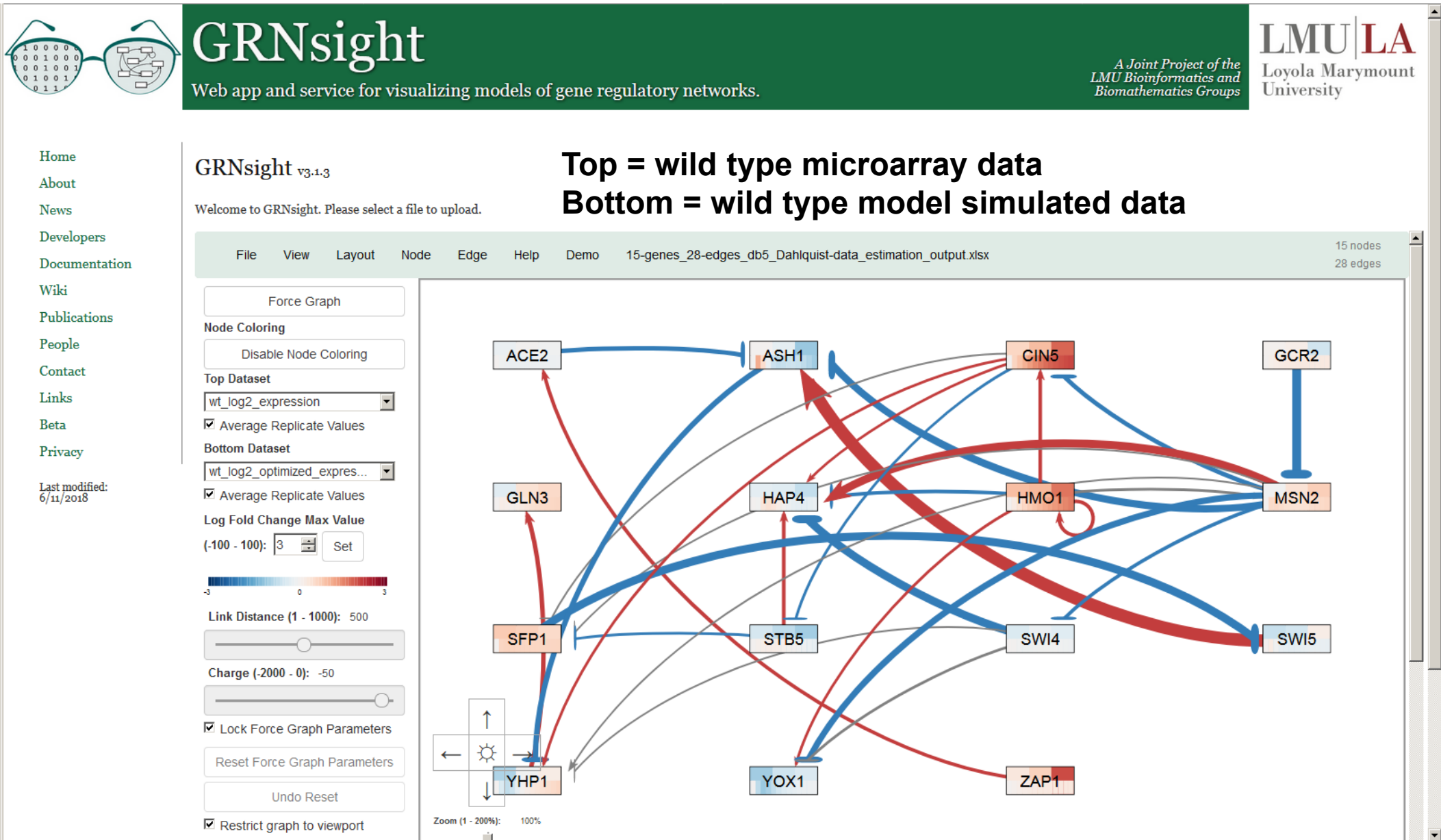


GRNsight automatically lays out GRNmap model results, displaying weight parameters



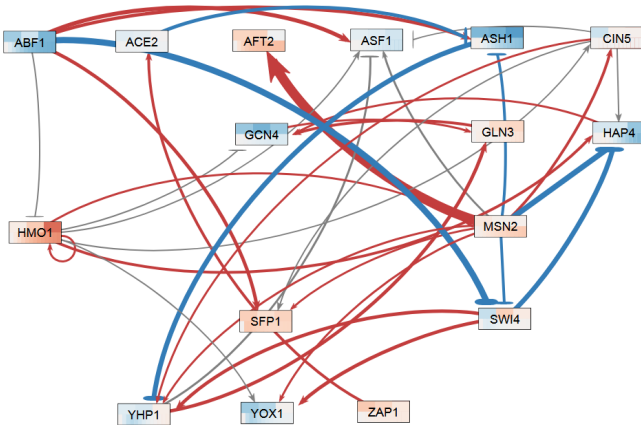
New in Version 2

Generally, we can see a good concordance between the measured and simulated expression data

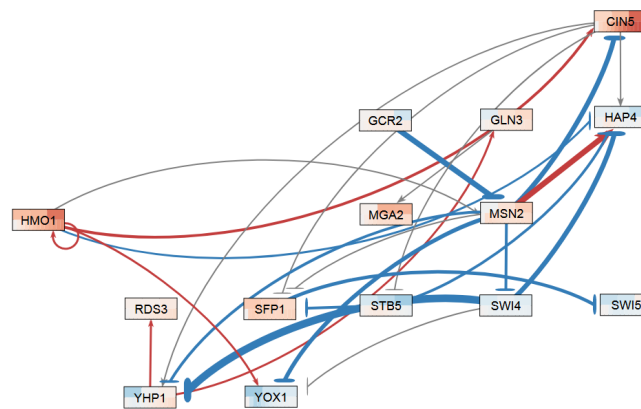


LSE:minLSE ratio used to determine relative model performance

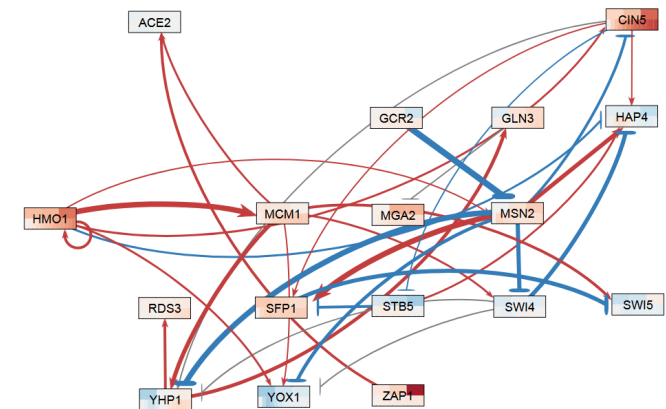
db1: 16 Genes, 26 Edges
LSE:minLSE = 1.42



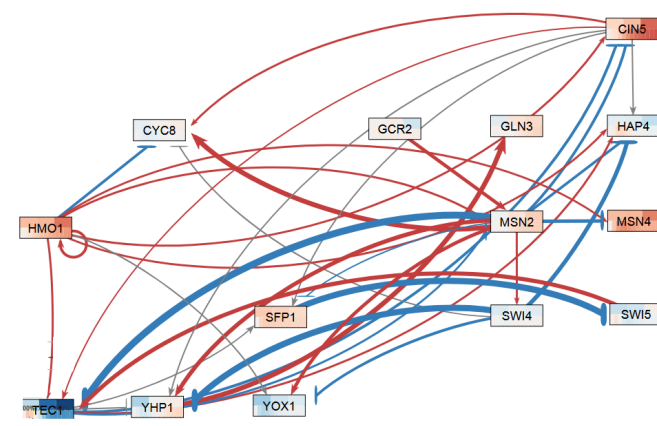
db2: 14 Genes, 25 Edges
LSE:minLSE = 1.36



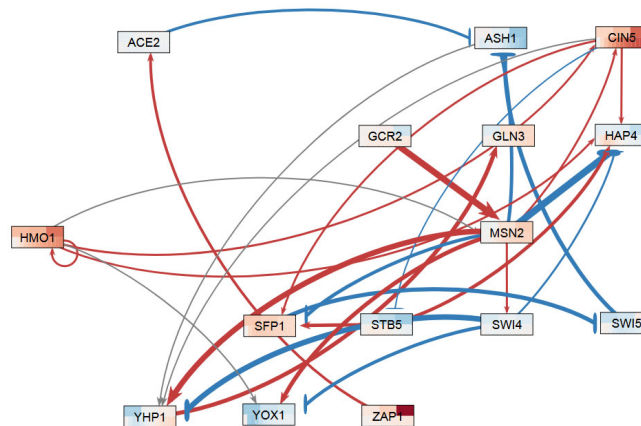
db3: 17 Genes, 32 Edges
LSE:minLSE = 1.41



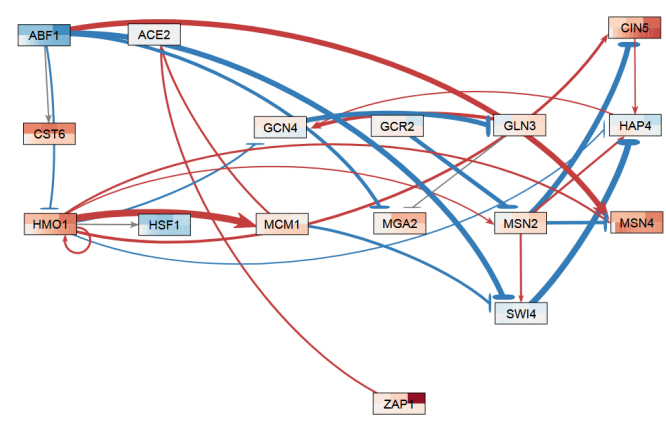
db4: 14 Genes, 35 Edges
LSE:minLSE = 1.30



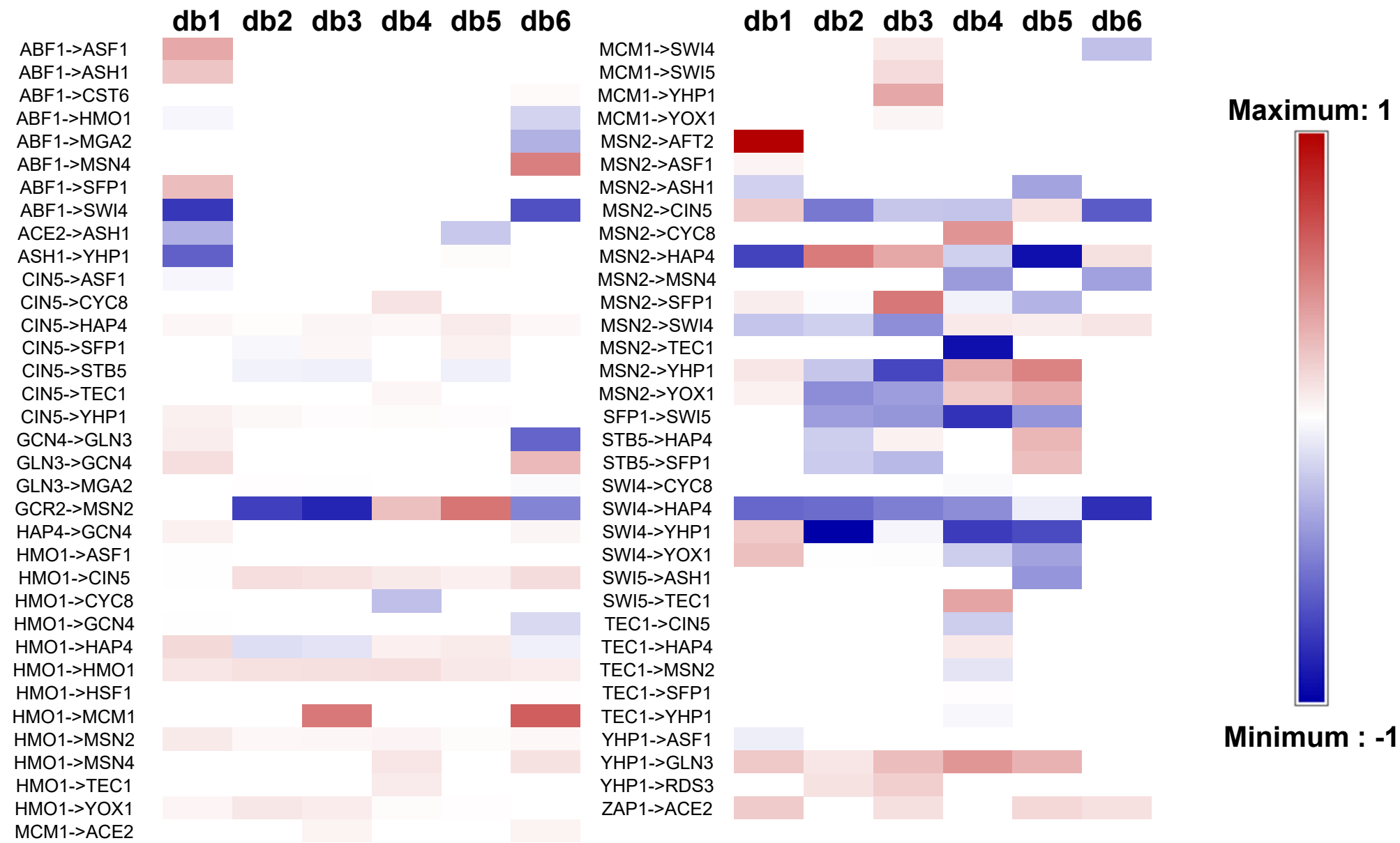
db5: 15 Genes, 28 Edges
LSE:minLSE = 1.43



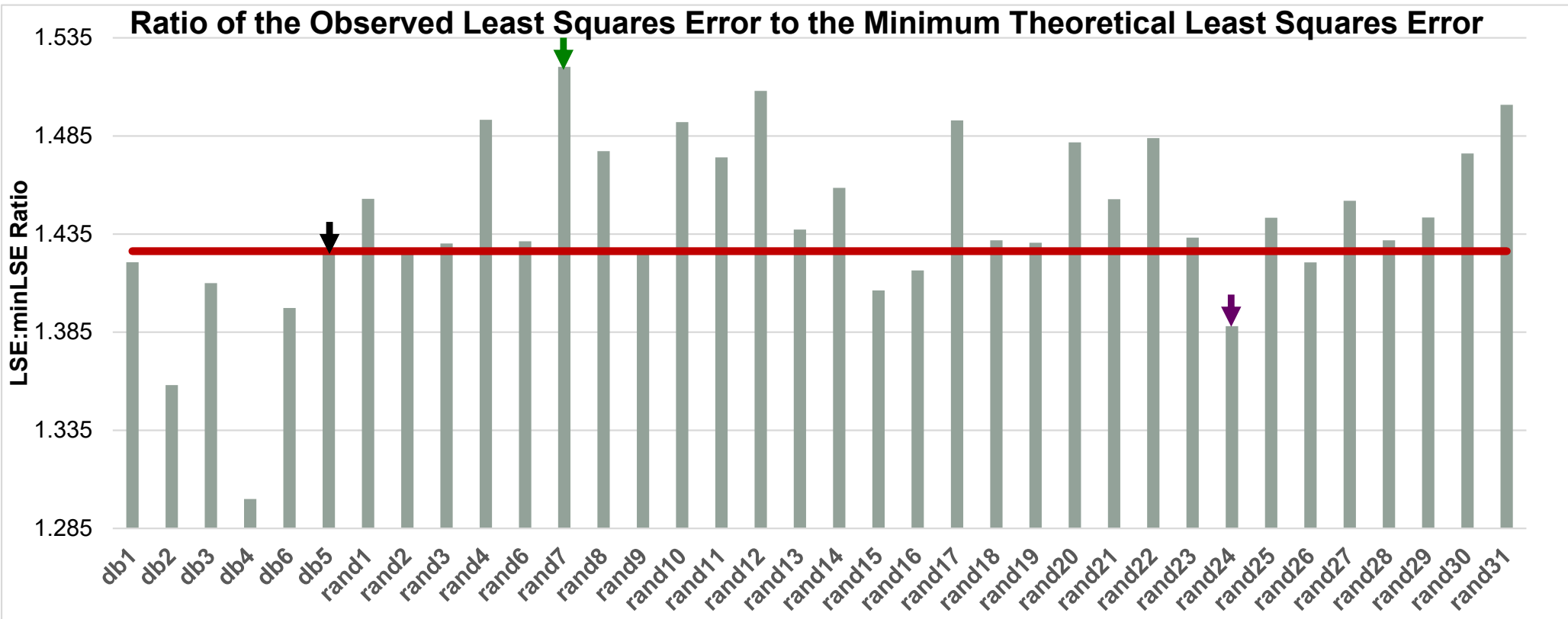
db6: 16 Genes, 27 Edges
LSE:minLSE = 1.40



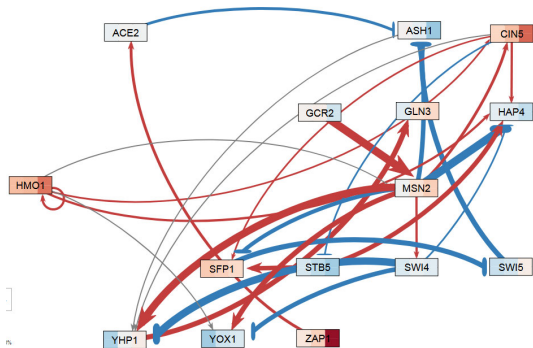
Modeling revealed consistent regulatory relationships among sets of transcription factors



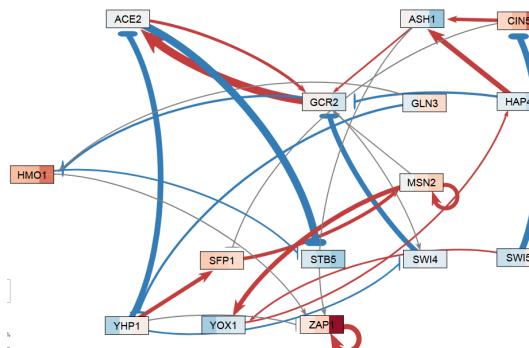
DB5 outperforms 30 random networks containing the same genes and numbers of edges



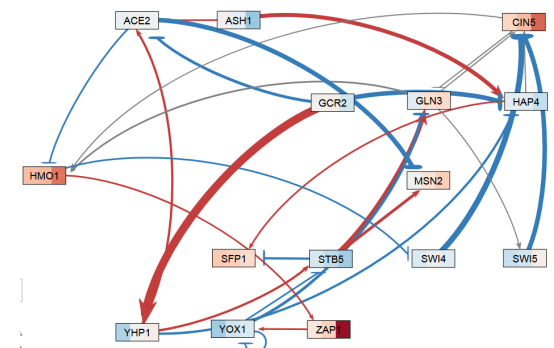
db5



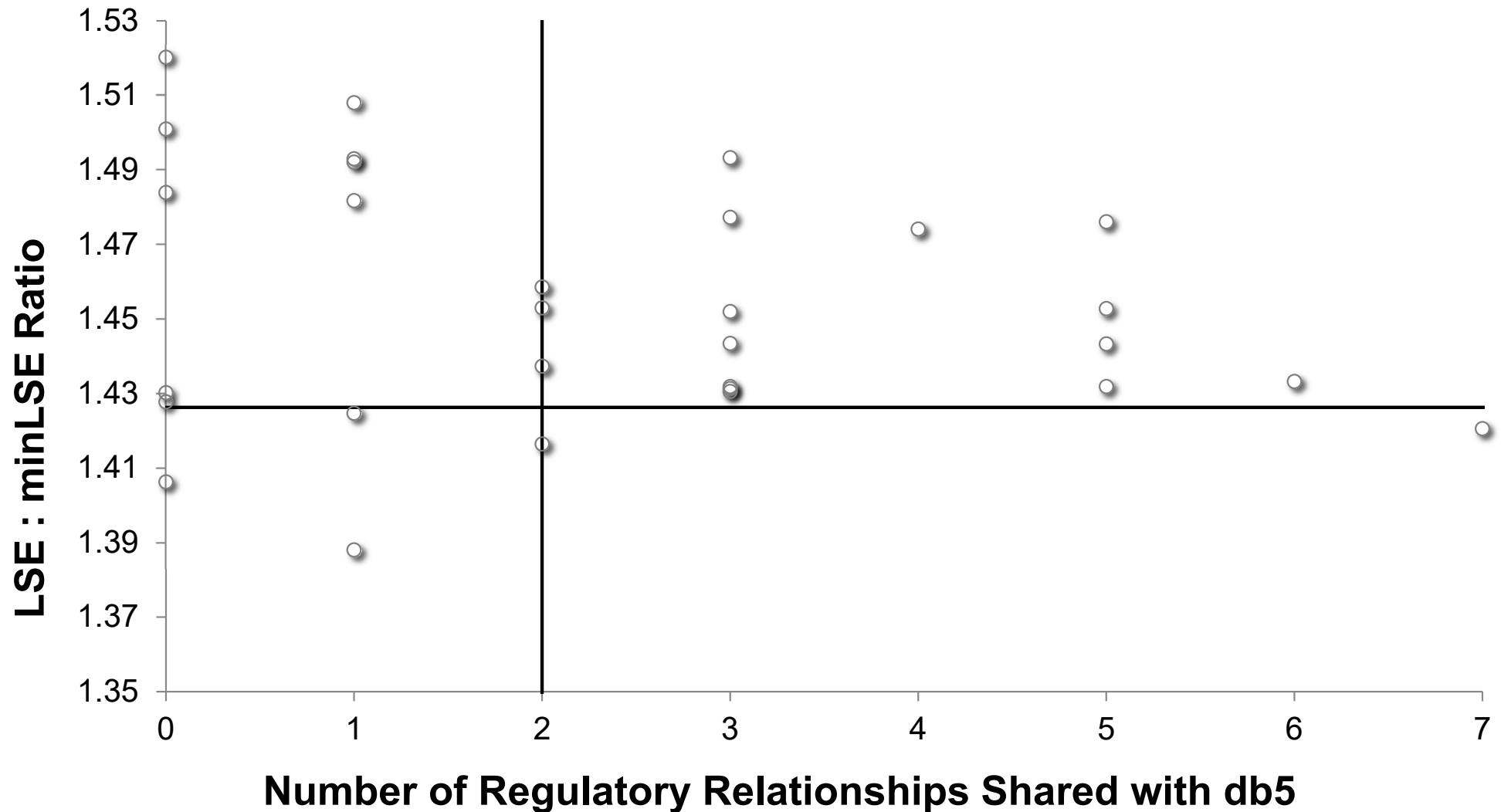
rand24



rand7

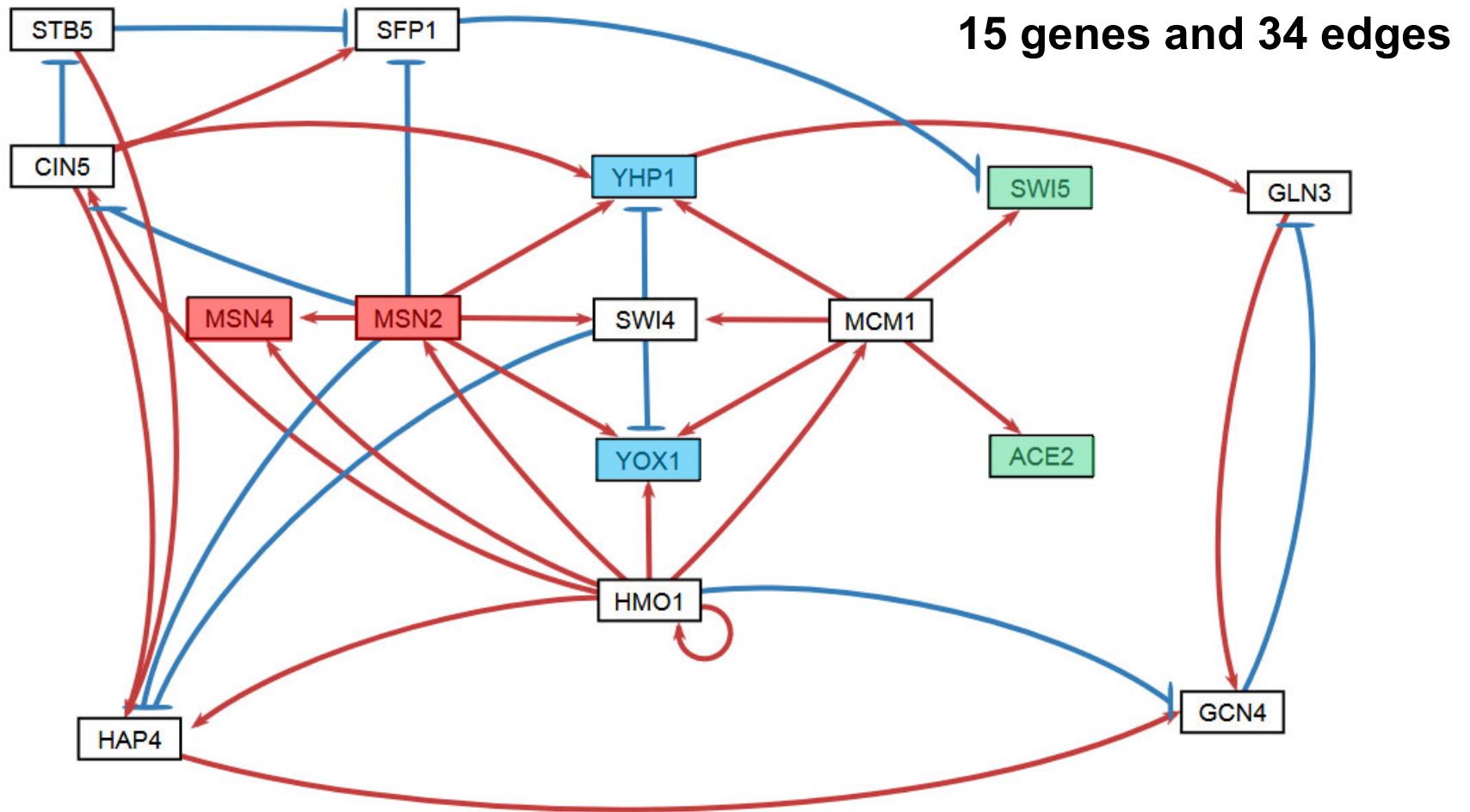


In general, random networks that share more connections with DB5 perform better



- Horizontal Line: LSE:minLSE ratio of db5 (1.4263).
- Vertical Line: median number of regulatory relationships shared with db5 (2).

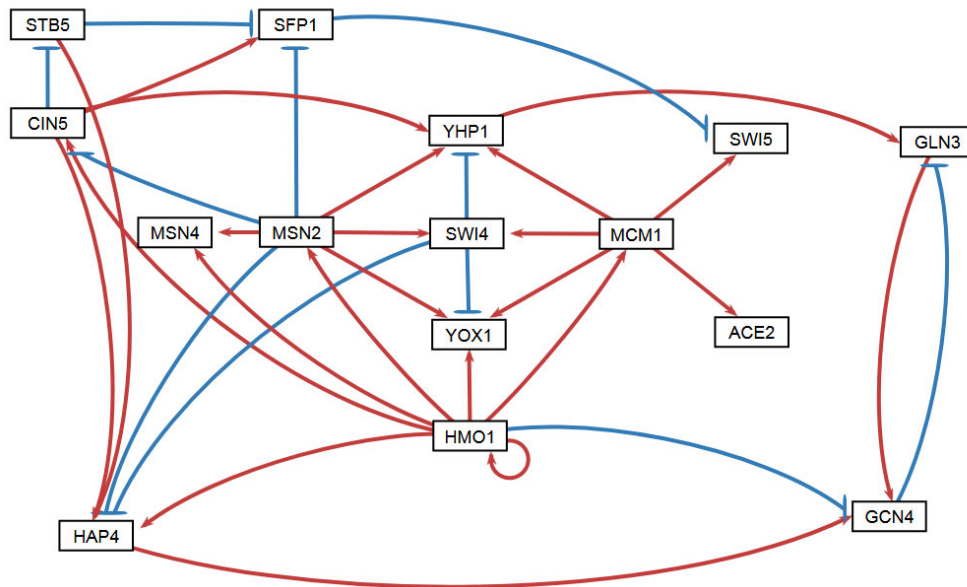
A hypothesis network was constructed and contains key regulators that are also paralog pairs



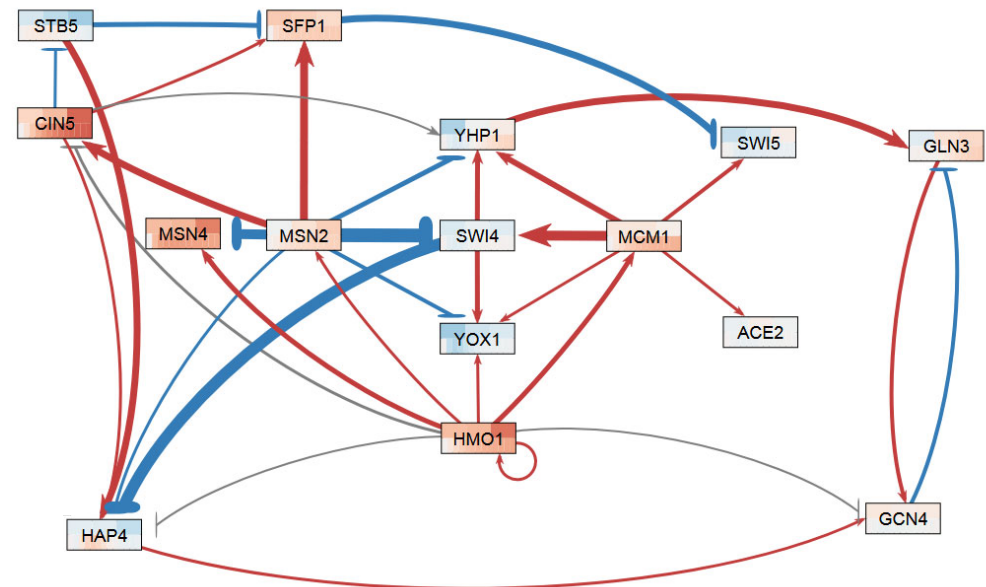
- **ACE2/SWI5 and YHP1/YOX1 regulate the cell cycle.**
- **MSN2/MSN4 regulate the general environmental stress response, repressing ribosome biogenesis.**
- **HMO1 induces TORC1-dependent upregulation of ribosome biogenesis .**

LSE:minLSE from the hypothesis network is smaller than 5 of 6 database-derived networks

Hypothesis

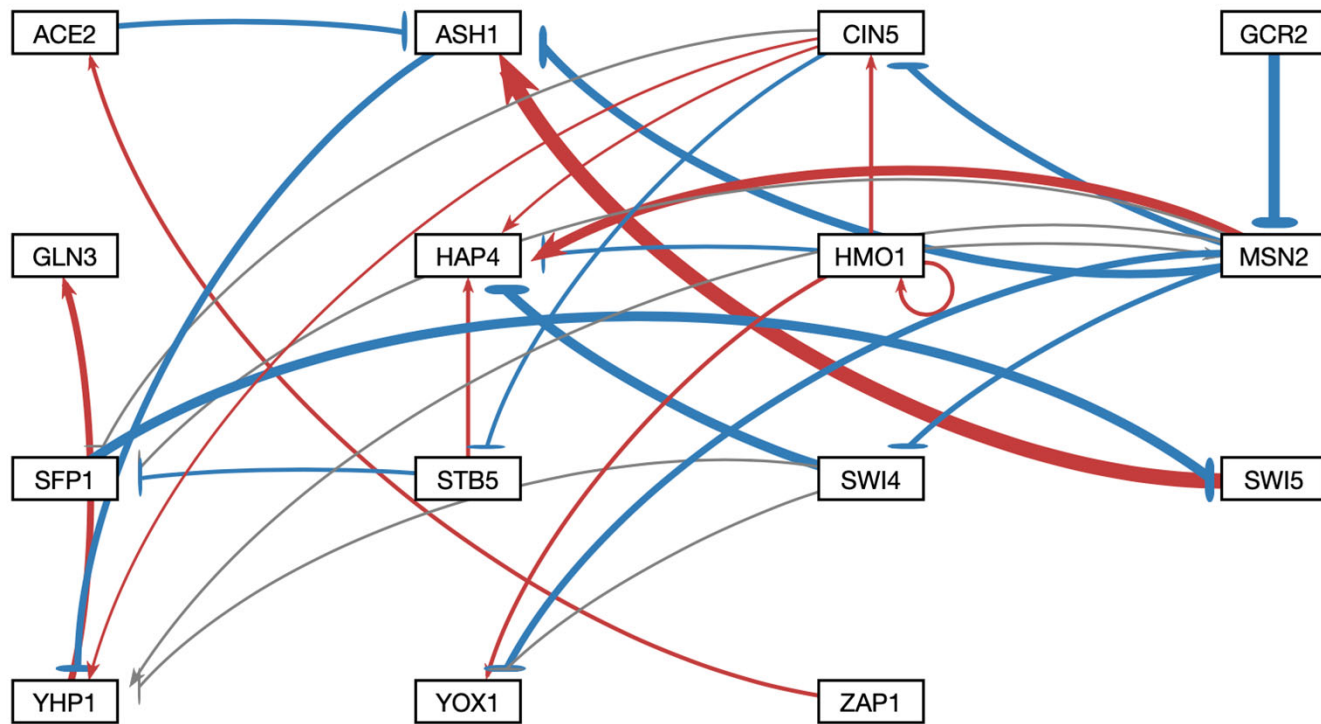


Modeling Results



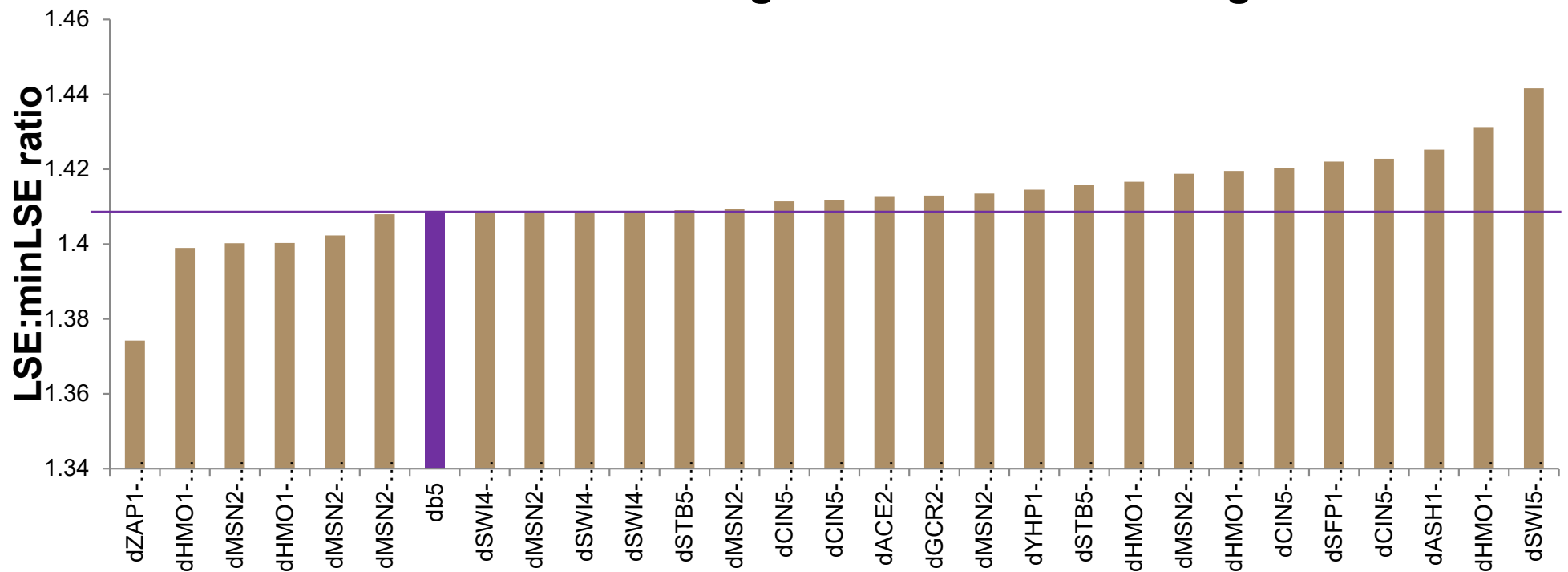
- **LSE:minLSE = 1.34**
- **25 out of 34 (74%) activation or repression relationships predicted correctly.**

Another approach was to systematically delete each of the 28 edges and re-estimate the parameters

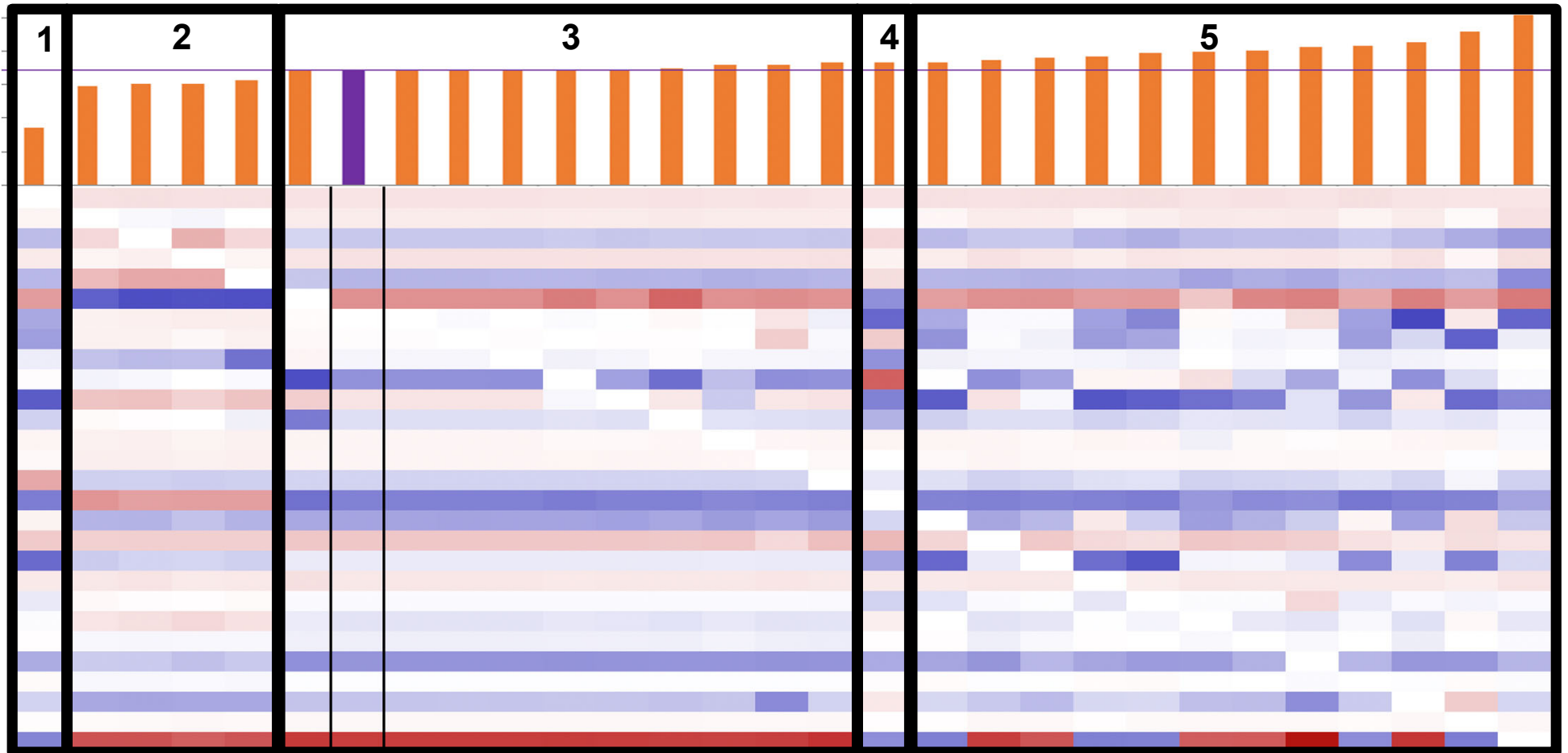


LSE:minLSE ratio compared for 28-edge deletion networks

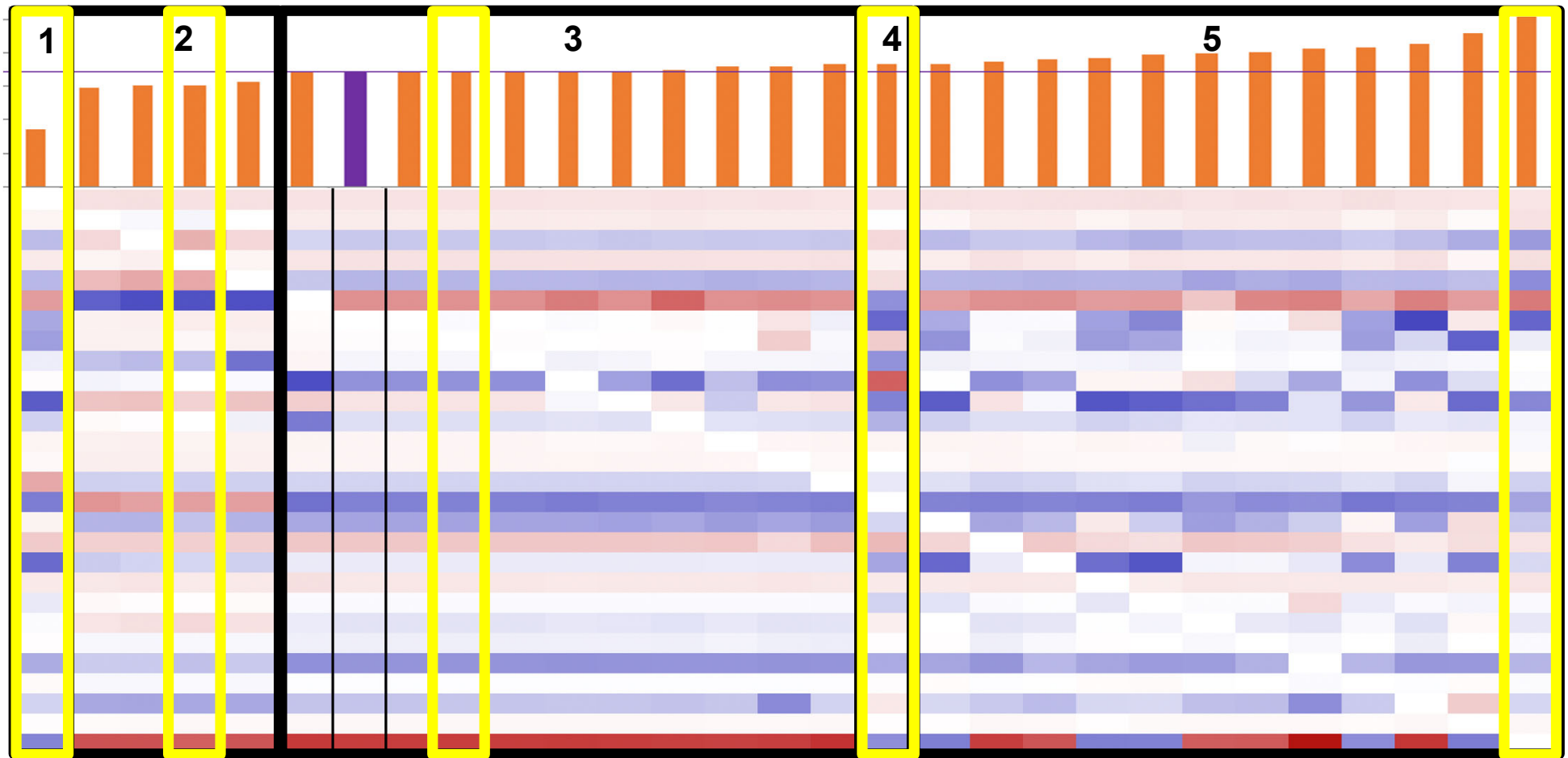
LSE:minLSE ratios arranged from smallest to largest



A heatmap of the weight parameters reveals 5 major groupings



One network from each group was selected for further analysis

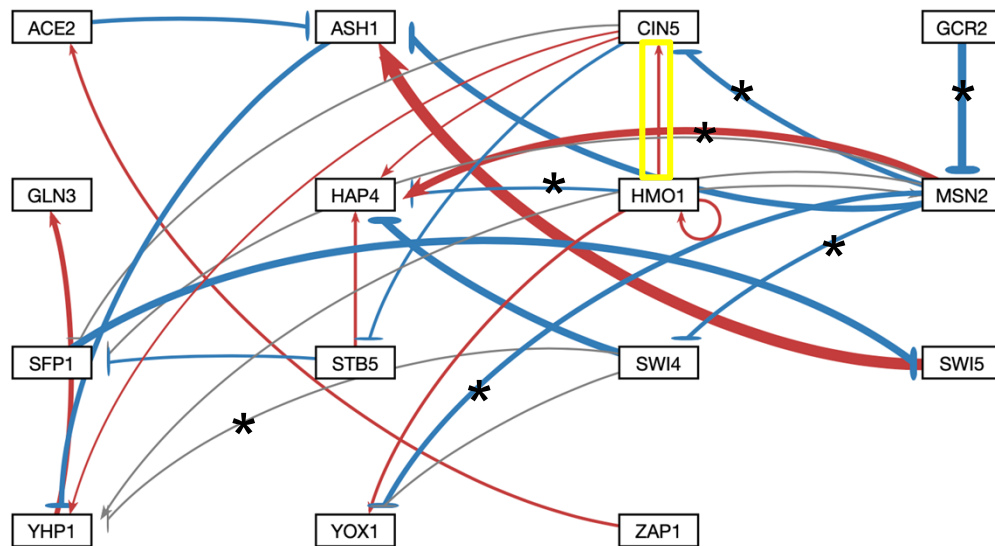


- 1) Edge ZAP1→ACE2 deleted
- 2) Edge HMO1→CIN5 deleted
- 3) Edge MSN2→YHP1 deleted

- 4) Edge GCR2→MSN2 deleted
- 5) Edge SWI5→ASH1 deleted

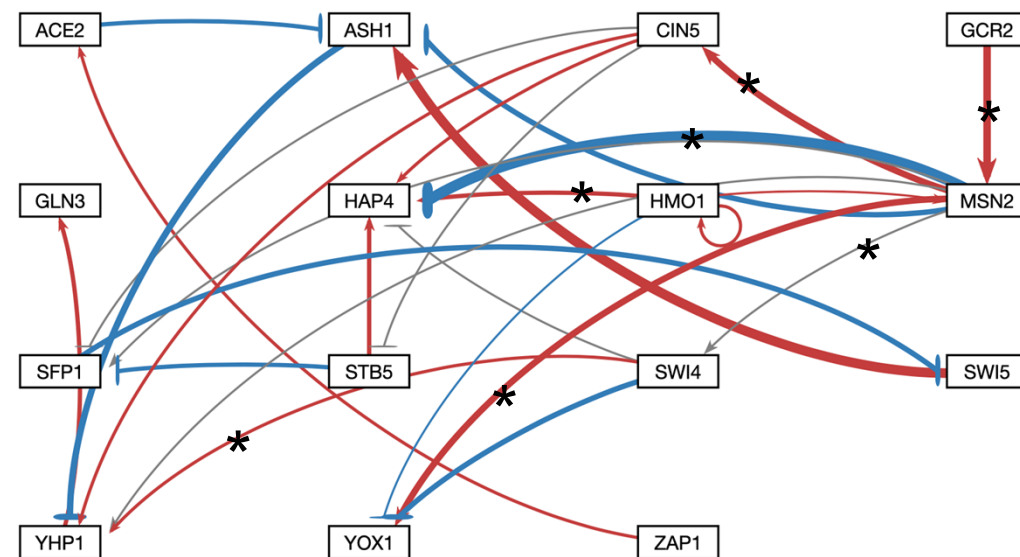
HMO1, CIN5, and MSN2 are central hubs in the network

Intact Network



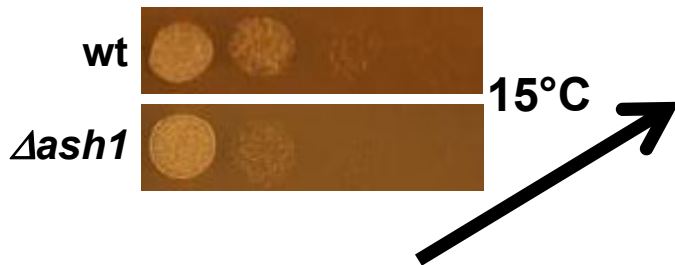
LSE:minLSE = 1.41

Group 2: Edge HMO1→CIN5 deleted

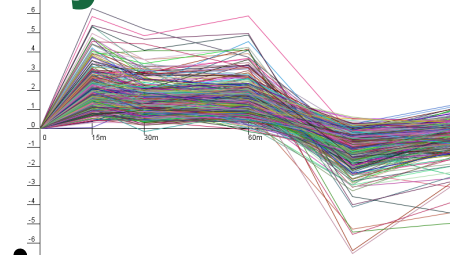
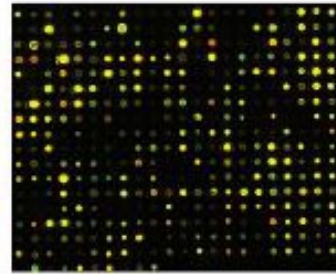


LSE:minLSE = 1.40

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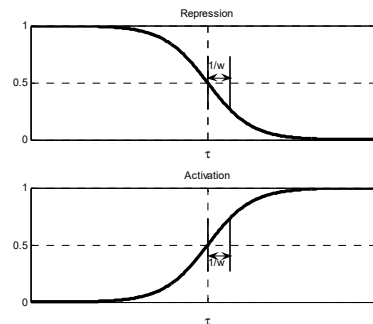
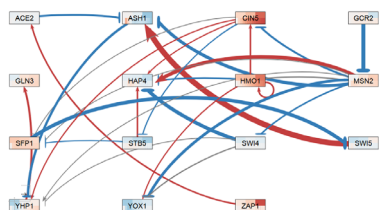
Cold shock microarray data from wt and TF deletion strains



Interpretation,
new questions,
new experiments

Normalization,
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Visualization of
modeling results
using GRNsight



Dynamical systems
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Derivation of gene
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Conclusions

- 1. Which transcription factors control the early response to cold shock?**
 - Experimental and modeling results suggest Cin5, Gln3, Hap4, Hmo1, Msn2, and Swi4 play important roles.
- 2. What are their relative levels of influence?**
 - Expression of target genes is regulated by a balance of activating and repressing transcription factors.
- 3. What are the indirect effects of other transcription factors in the gene regulatory network?**
 - Different approaches to analysis have highlighted the importance of certain edges in the network.

GRNsight: <http://dondi.github.io/GRNsight/>
GRNmap: <http://kdahlquist.github.io/GRNmap>

Ben G. Fitzpatrick (LMU Mathematics)
John David N. Dionisio (LMU Computer Science)

Current students:

Nikki C. Chun (Biology '26)
A'Kaia Phelps (Computer Science '26)
Ngoc K. Tran (Computer Science '26)
Cecilia J. Zaragoza (Computer Science '26)

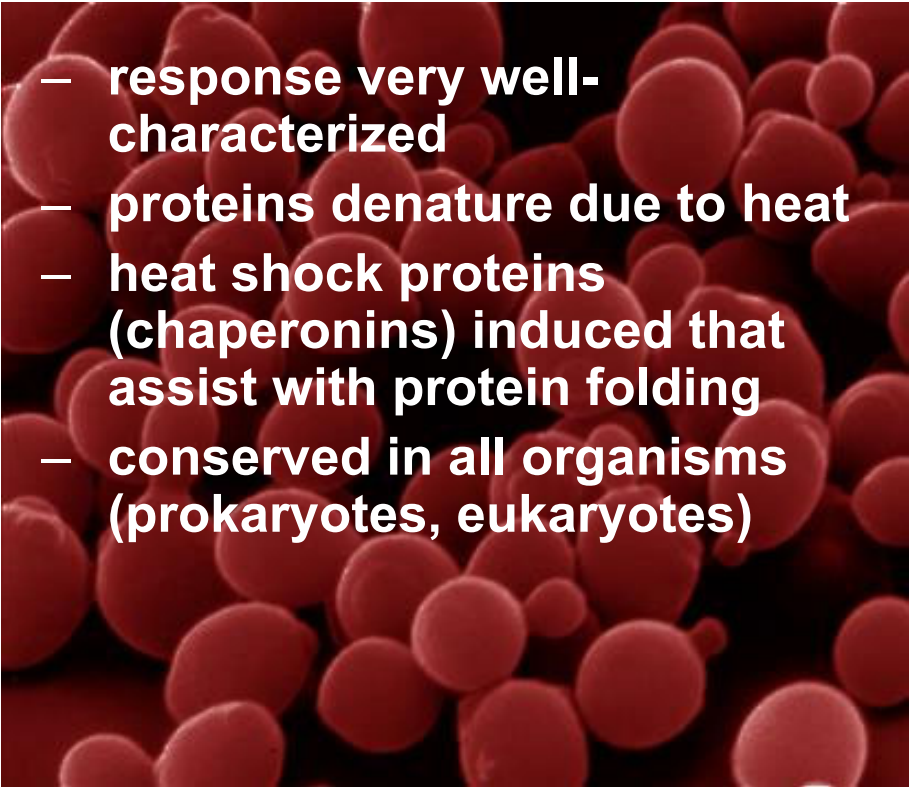
Funding

NSF RUI, Kadner-Pitts Research Grant, SURP, Honors Program, Rains Research Assistant Program

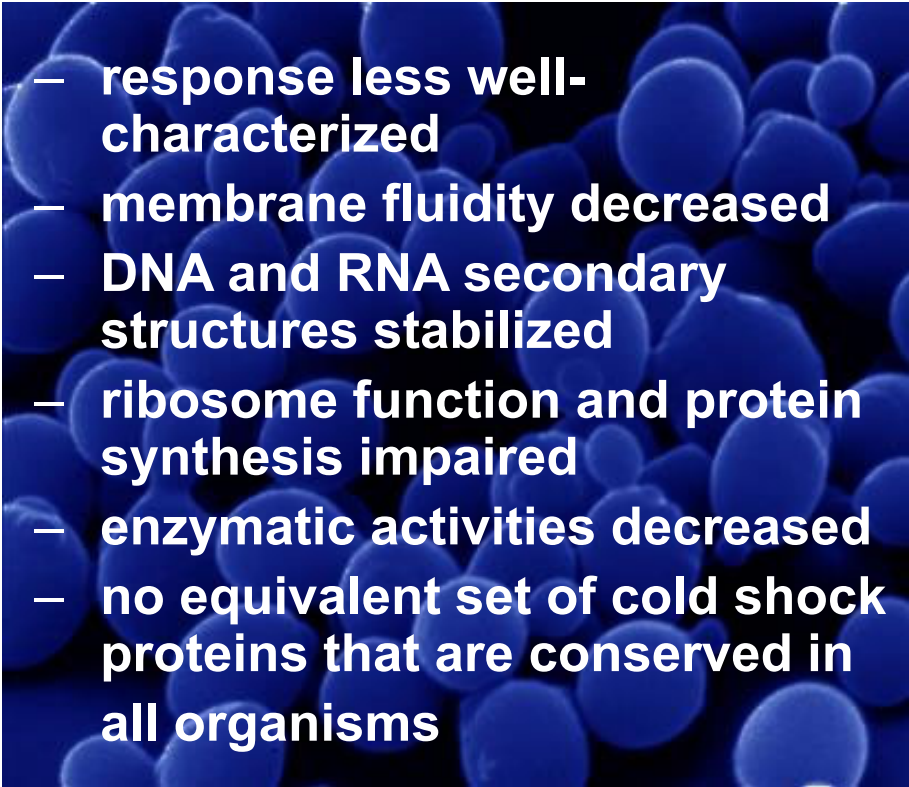


Cold shock is an environmental stress that is not well-studied

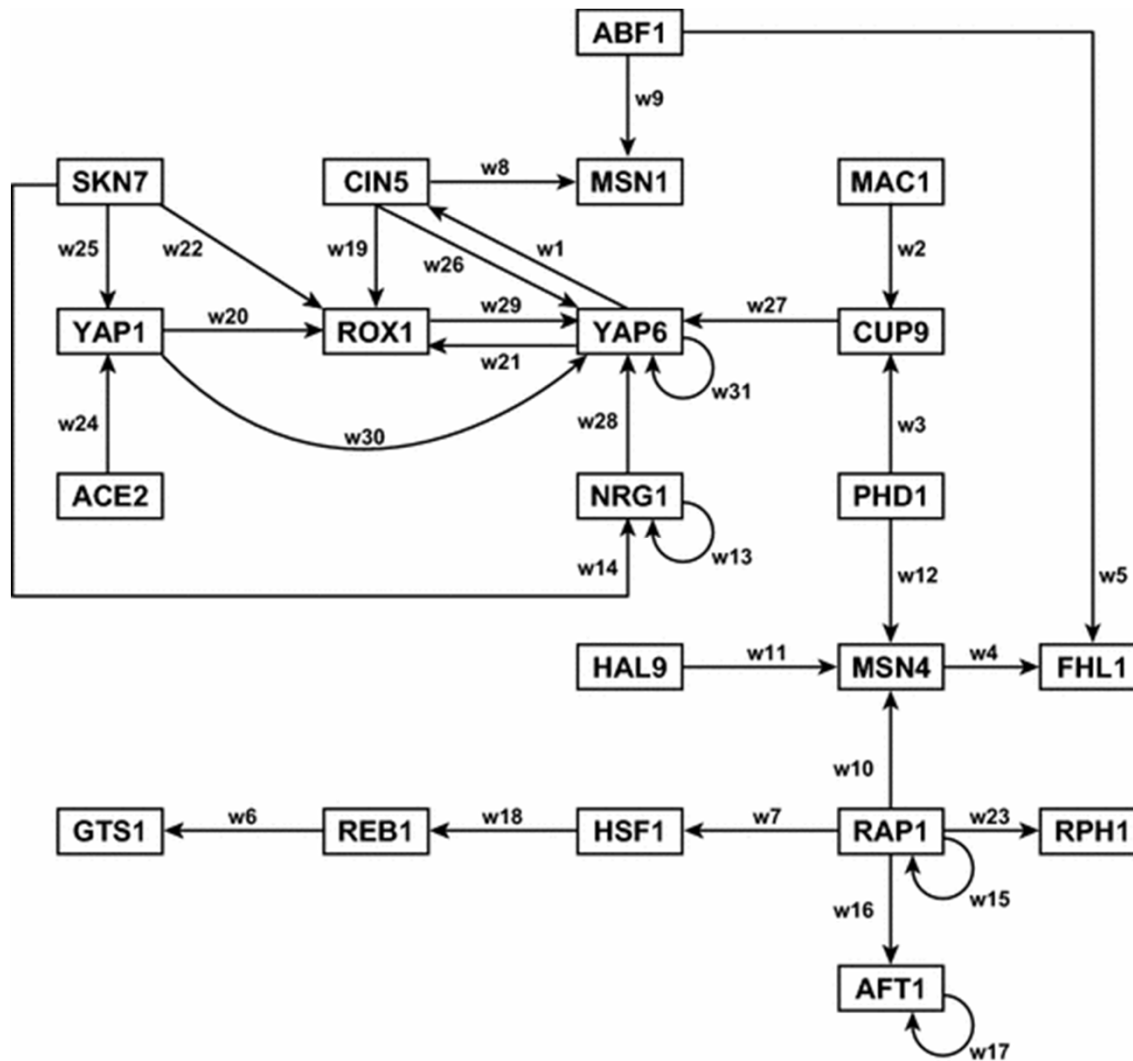
Heat shock

- 
- response very well-characterized
 - proteins denature due to heat
 - heat shock proteins (chaperonins) induced that assist with protein folding
 - conserved in all organisms (prokaryotes, eukaryotes)

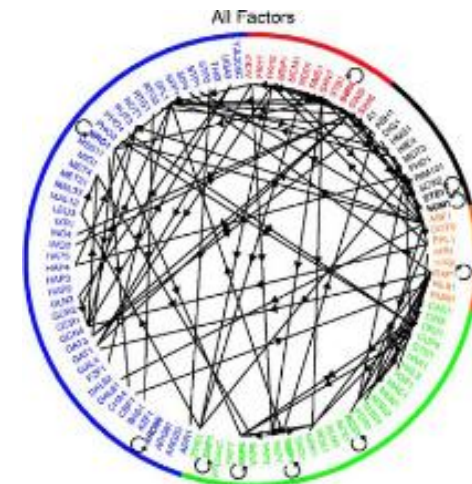
Cold shock

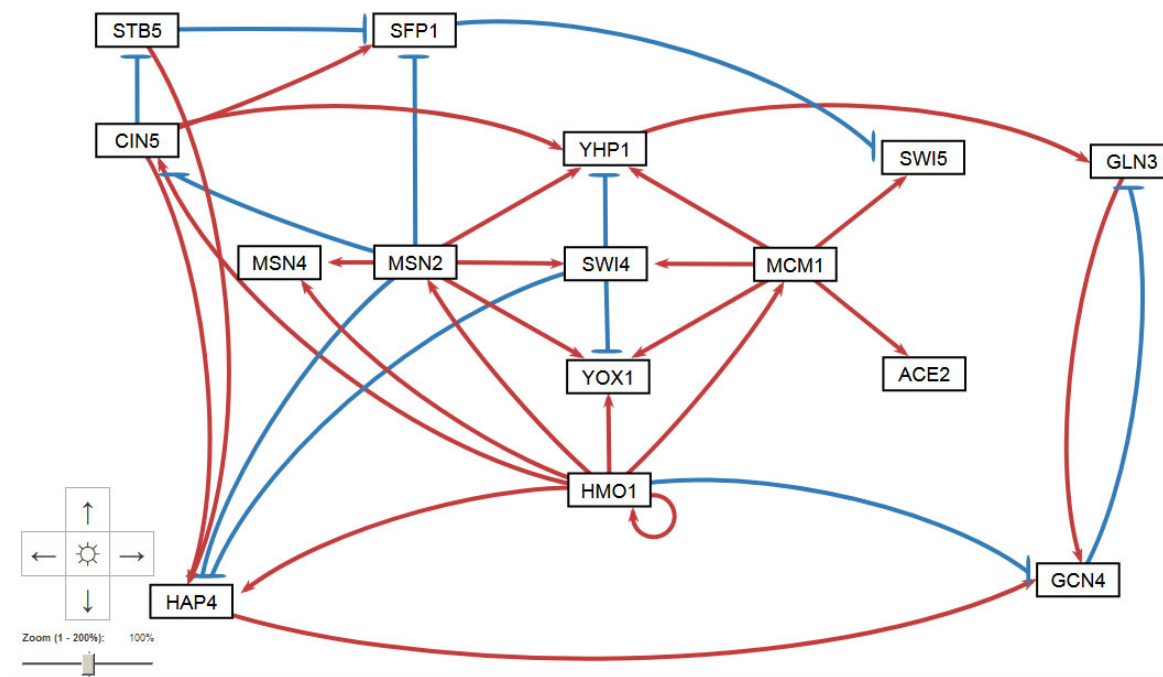
- 
- response less well-characterized
 - membrane fluidity decreased
 - DNA and RNA secondary structures stabilized
 - ribosome function and protein synthesis impaired
 - enzymatic activities decreased
 - no equivalent set of cold shock proteins that are conserved in all organisms

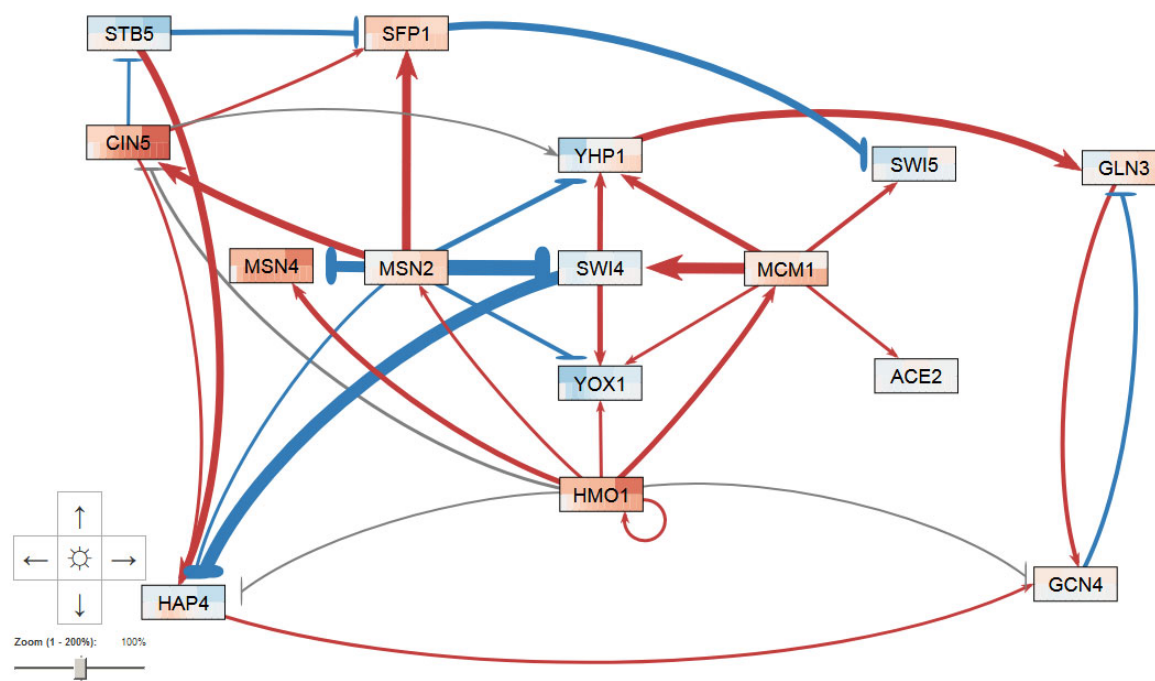
A First Pass at a “Medium-Scale” Gene Regulatory Network that Regulates the Cold Shock Response



- Each node represents one gene encoding one transcription factor.
- Each edge represents a regulatory relationship, either activation or repression.
- Network structure based on Lee et al. (2002) data.

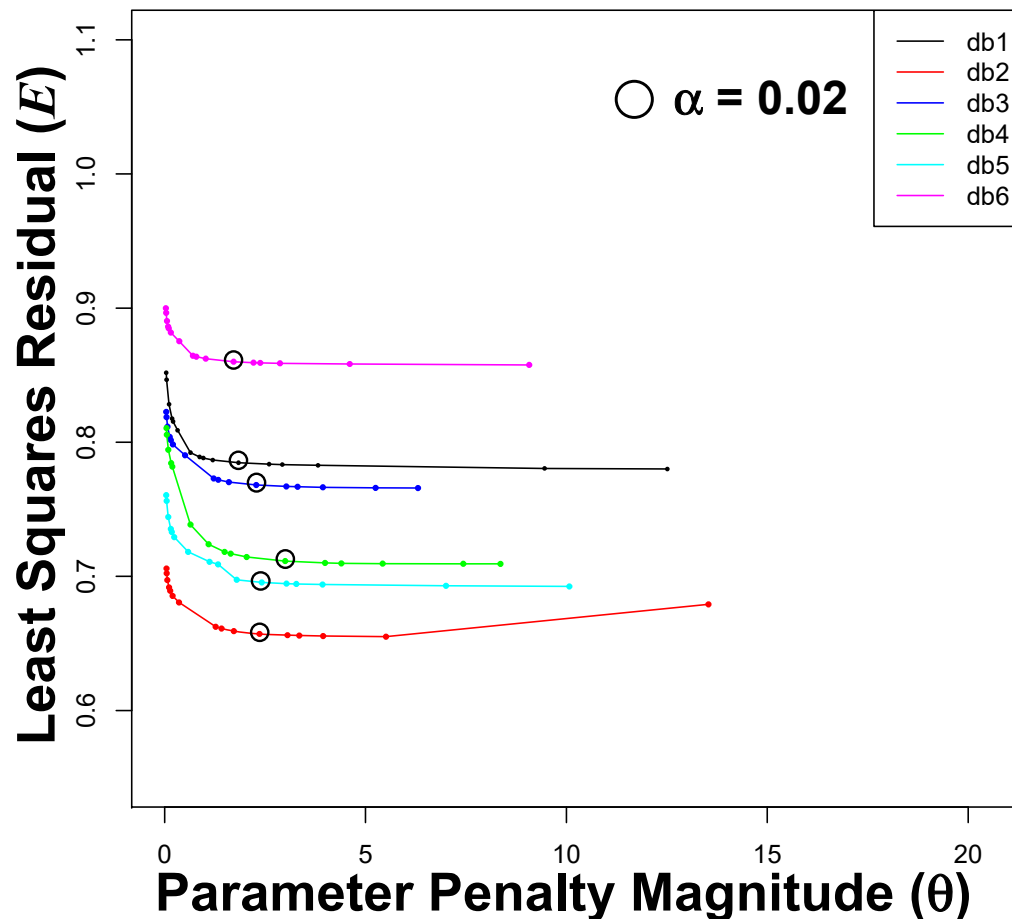






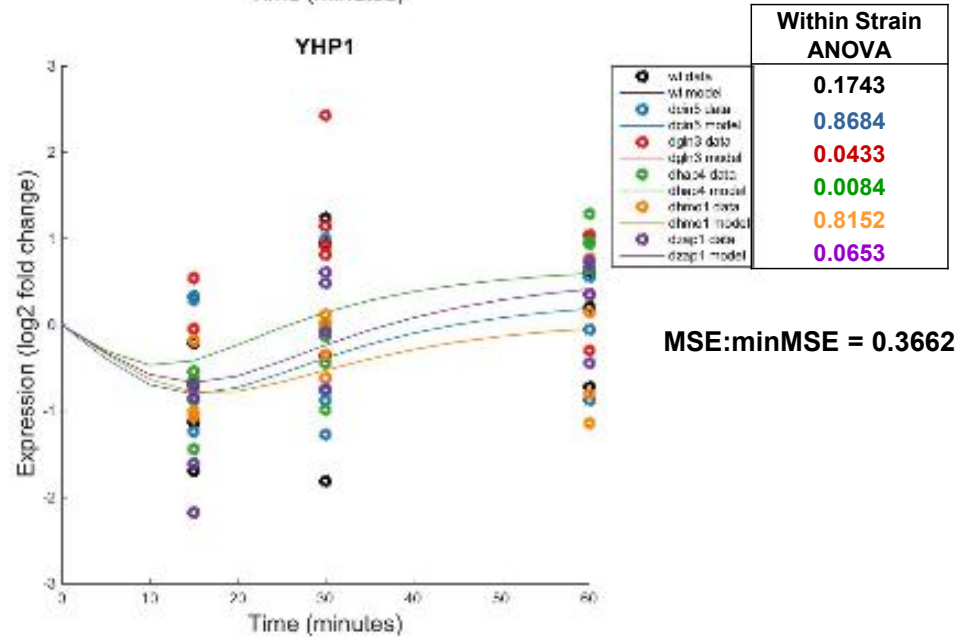
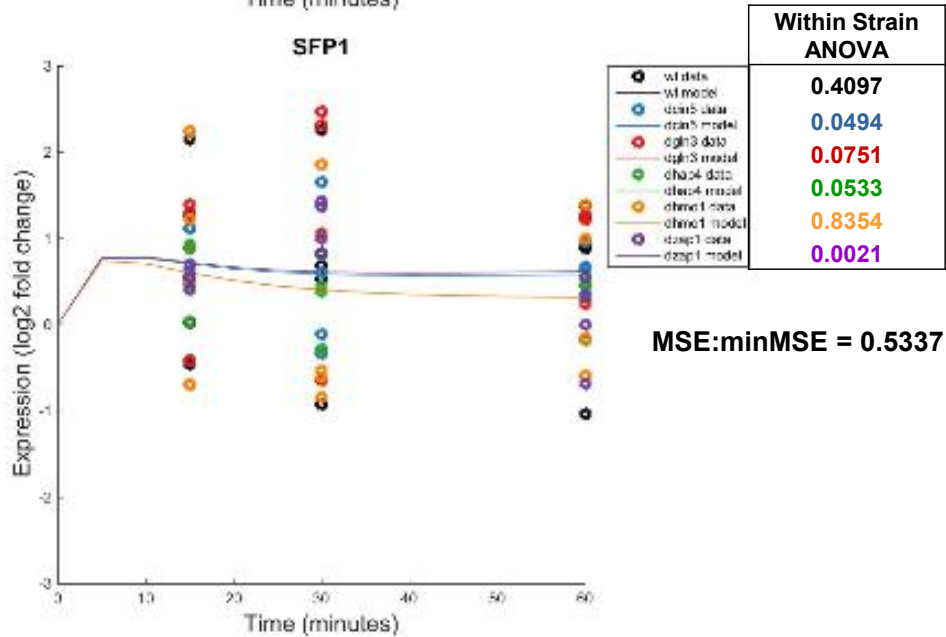
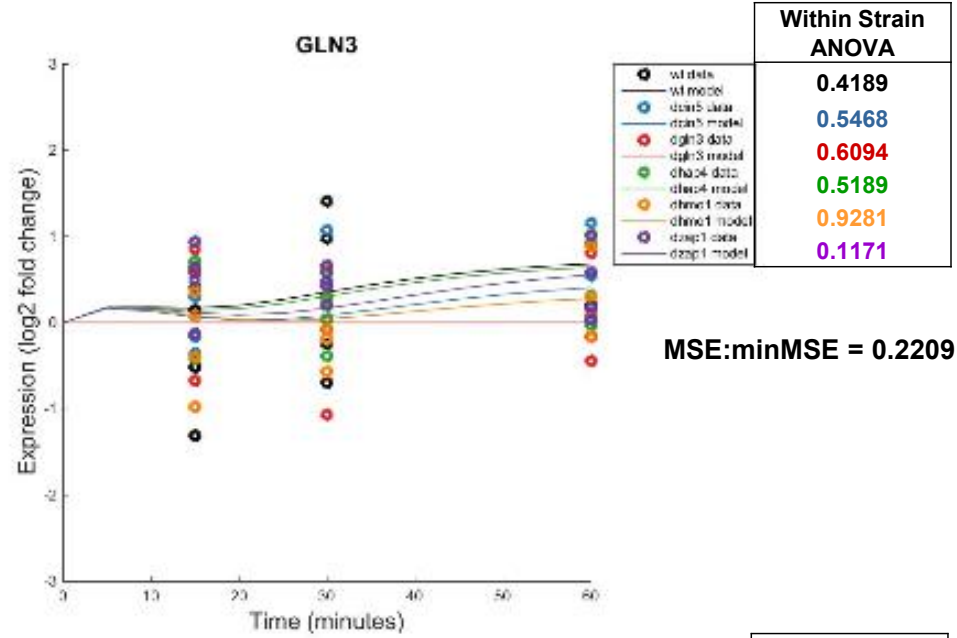
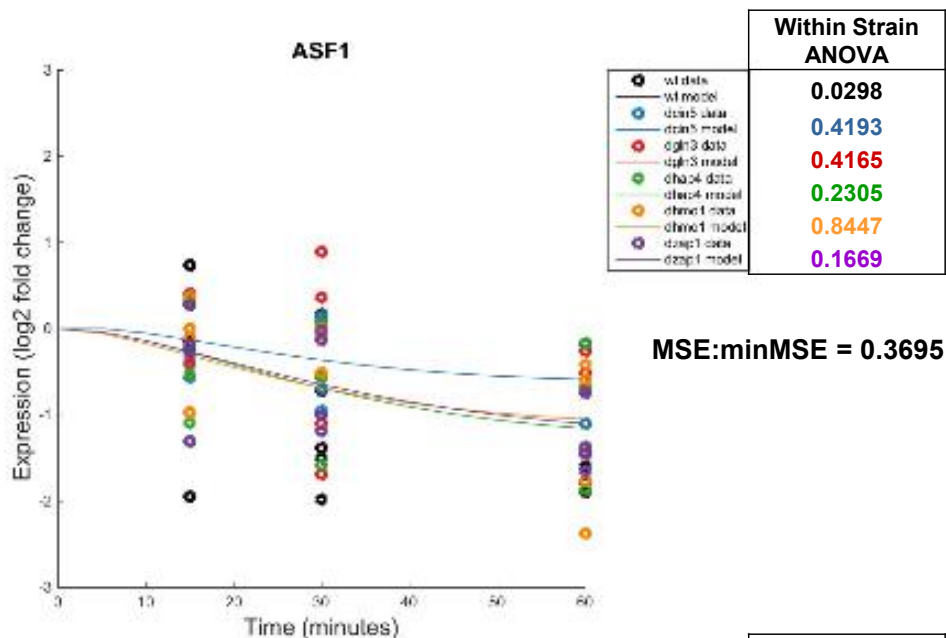
An Alpha Value of 0.02 Is Also Reasonable for DB1-DB6

$$E = \frac{1}{n_g n_t} \sum_{i=1}^{n_g} \sum_{k=1}^{n_t} [z^d(t_r) - z^c(t_r)]^2 + \alpha \|\theta\|^2$$



- Data for wt and 5 deletion strains are used.
- We also experimented with larger networks of up to 35 genes.
- This penalized least squares approach is really good at getting the parameters to minimize...
- ...but, we doubted the results and our ability to interpret them when networks got much larger than ~15 genes.

Forward Simulation of the Models Fits the Microarray Data Well (Note that data are noisy!)



The GRNmap Mathematical Model Answers the Questions:

2. What are the relative levels of influence of each TF?
3. What are the indirect effects of other transcription factors in the gene regulatory network?

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	cols regulators/rows targets	ABF1	ACE2	AFT1	CIN5	CUP9	FHL1	GTS1	HAL9	HSF1	MAC1	MSN1	MSN4	NRG1	PHD1	RAP1	REB1	ROX1	RPH1	SKN7	YAP1	YAP6
2	ABF1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	ACE2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	AFT1	0	0	-0.8966	0	0	0	0	0	0	0	0	0	0	0	-0.4030	0	0	0	0	0	0
5	CIN5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.0450
6	CUP9	0	0	0	0	0	0	0	0	0	-0.1882	0	0	0	-0.6510	0	0	0	0	0	0	0
7	FHL1	0.1562	0	0	0	0	0	0	0	0	0	0	0.6121	0	0	0	0	0	0	0	0	0
8	GTS1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0778	0	0	0	0	0
9	HAL9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	HSF1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1.2321	0	0	0	0	0	0
11	MAC1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	MSN1	-2.9707	0	0	0.9393	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	MSN4	0	0	0	0	0	0	0	1.4283	0	0	0	0	0	0.5447	1.0131	0	0	0	0	0	0
14	NRG1	0	0	0	0	0	0	0	0	0	0	0	0	1.2341	0	0	0	0	0	-0.1852	0	0
15	PHD1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
16	RAP1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.8890	0	0	0	0	0	0
17	REB1	0	0	0	0	0	0	0	0	-0.0102	0	0	0	0	0	0	0	0	0	0	0	0
18	ROX1	0	0	0	-0.9278	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.5744	-0.4315	-0.5071
19	RPH1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.4999	0	0	0	0	0	0
20	SKN7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
21	YAP1	0	-1.3615	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.4082	0	0
22	YAP6	0	0	0	-0.5312	-0.1293	0	0	0	0	0	0	0	0.6215	0	0	0	-0.7503	0	0	0.0146	-0.3027

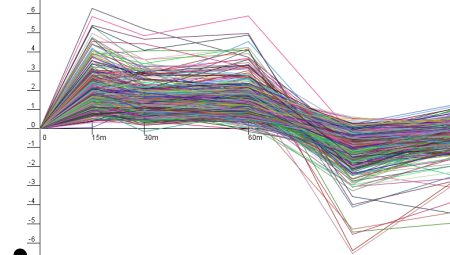
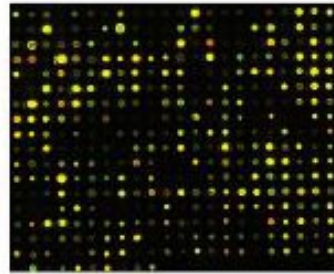
But how do we make sense of these weight values?

Systems Biology Approach to Understanding the Regulation of the Cold Shock Response in Yeast

Current GRNsight Coding Team:

- Eileen Choe
- Mihir Samdarshi
- Jen Shin
- BIOL 367: Biological Databases

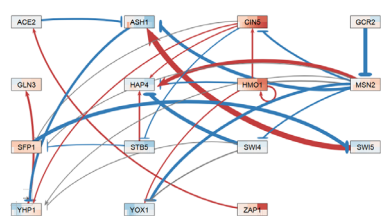
Cold shock microarray data from wt and TF deletion strains



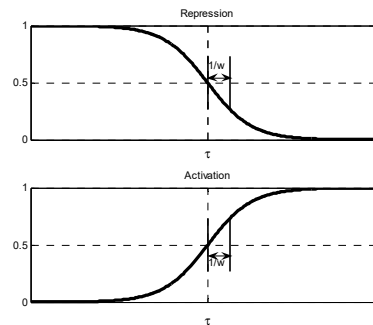
Normalization, statistical analysis, clustering

Derivation of gene regulatory networks from YEASTRACT

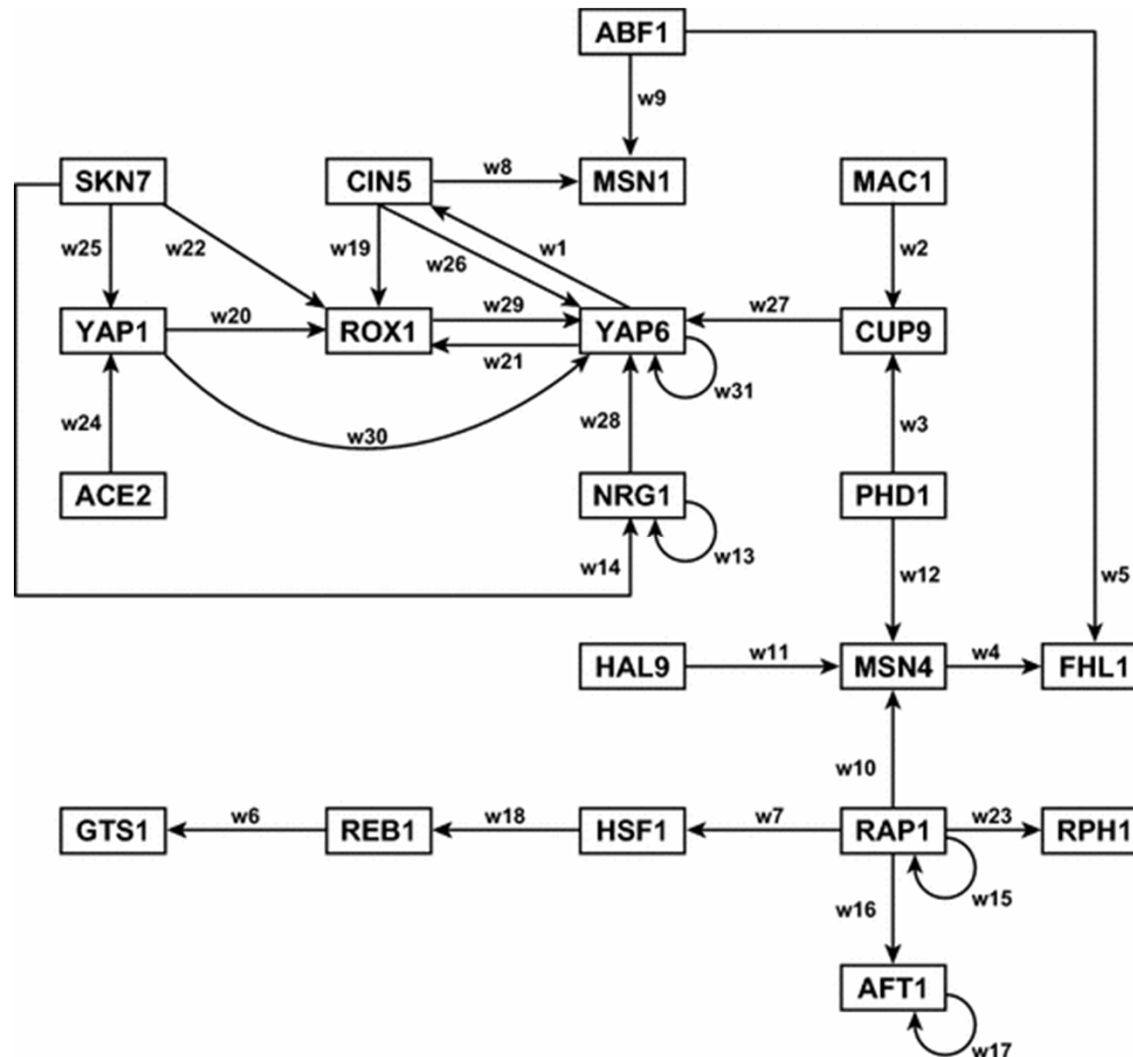
Visualization of modeling results using GRNsight



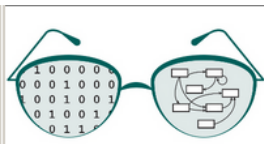
Dynamical systems modeling using GRNmap



Our “First Pass” GRN Diagram Took Several Hours to Create by Hand in Adobe Illustrator



GRNsight Rapidly Generates GRN Graphs Using Our Customizations to the Open Source D3 Library



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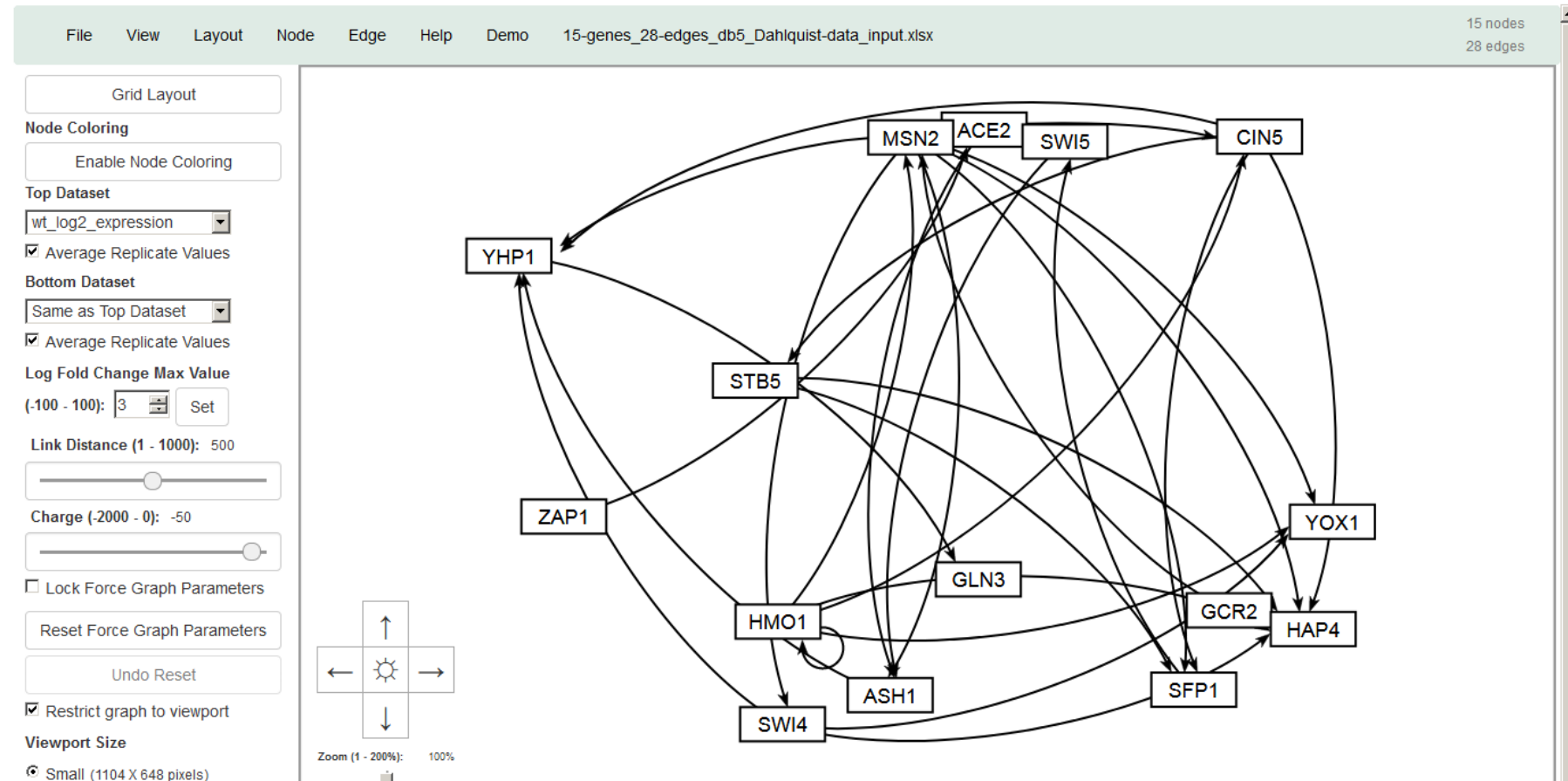
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6/11/2018

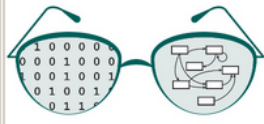
GRNsight v3.1.3

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Dahlquist et al. (2016) *PeerJ Computer Science*; Version 1 available March 2014

Besides the Force Graph Layout, Users Can Rearrange the Nodes by Hand



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GRNsight v3.1.3

Welcome to GRNsight. Please select a file to upload.

FileViewLayoutNodeEdgeHelpDemo15-genes_28-edges_db5_Dahlquist-data_input.xlsx15 nodes28 edges

Force Graph

Node Coloring

Enable Node Coloring

Top Dataset

wt_log2_expression

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value

(-100 - 100): 3Set

Link Distance (1 - 1000): 500

Charge (-2000 - 0): -50

☒ Lock Force Graph Parameters

Reset Force Graph Parameters

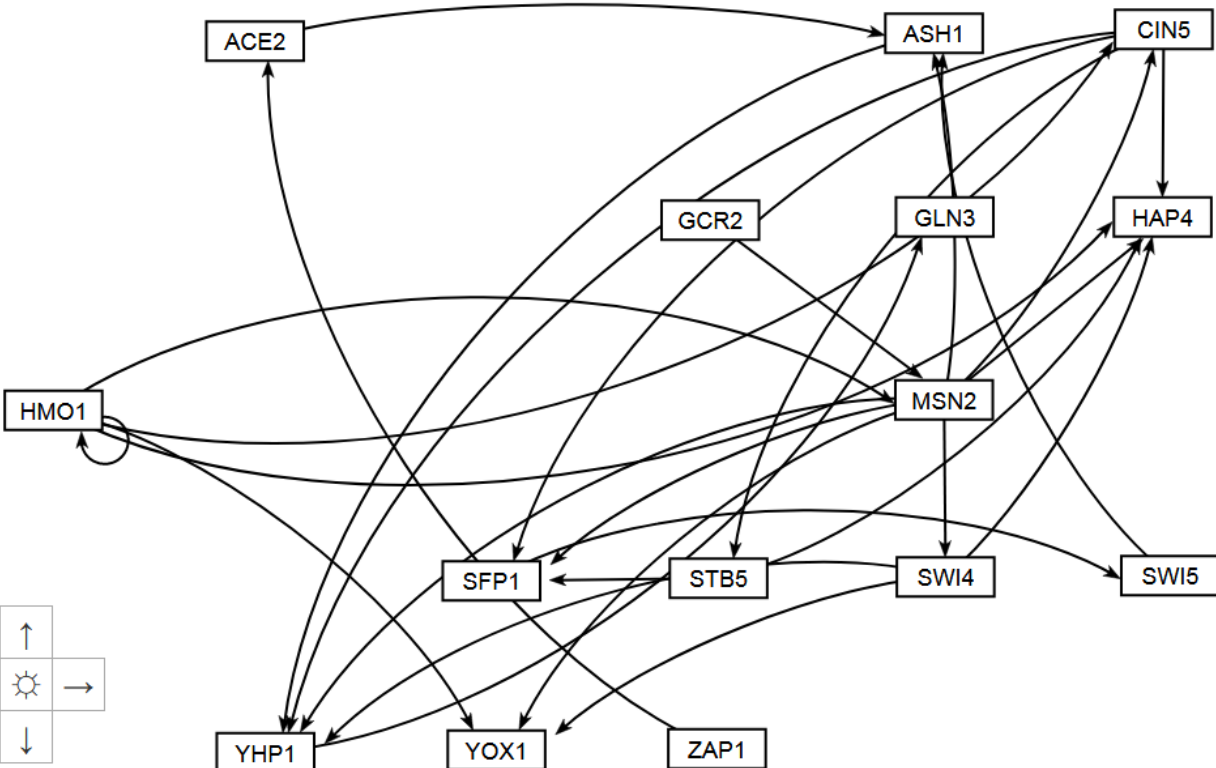
Undo Reset

☒ Restrict graph to viewport

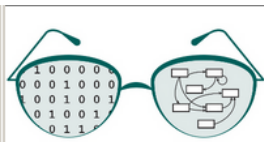
Viewport Size

Small (1104 X 648 pixels)

Zoom (1 - 200%): 100%



Or Choose the Convenient Block Layout Option



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GRNsight v3.1.3

Welcome to GRNsight. Please select a file to upload.

File View Layout Node Edge Help Demo 15-genes_28-edges_db5_Dahlquist-data_input.xlsx

15 nodes
28 edges

Force Graph

Node Coloring

Enable Node Coloring

Top Dataset

wt_log2_expression

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value

(-100 - 100): 3

Set

Link Distance (1 - 1000): 500

Charge (-2000 - 0): -50

Link Distance (1 - 1000): 500

☒ Lock Force Graph Parameters

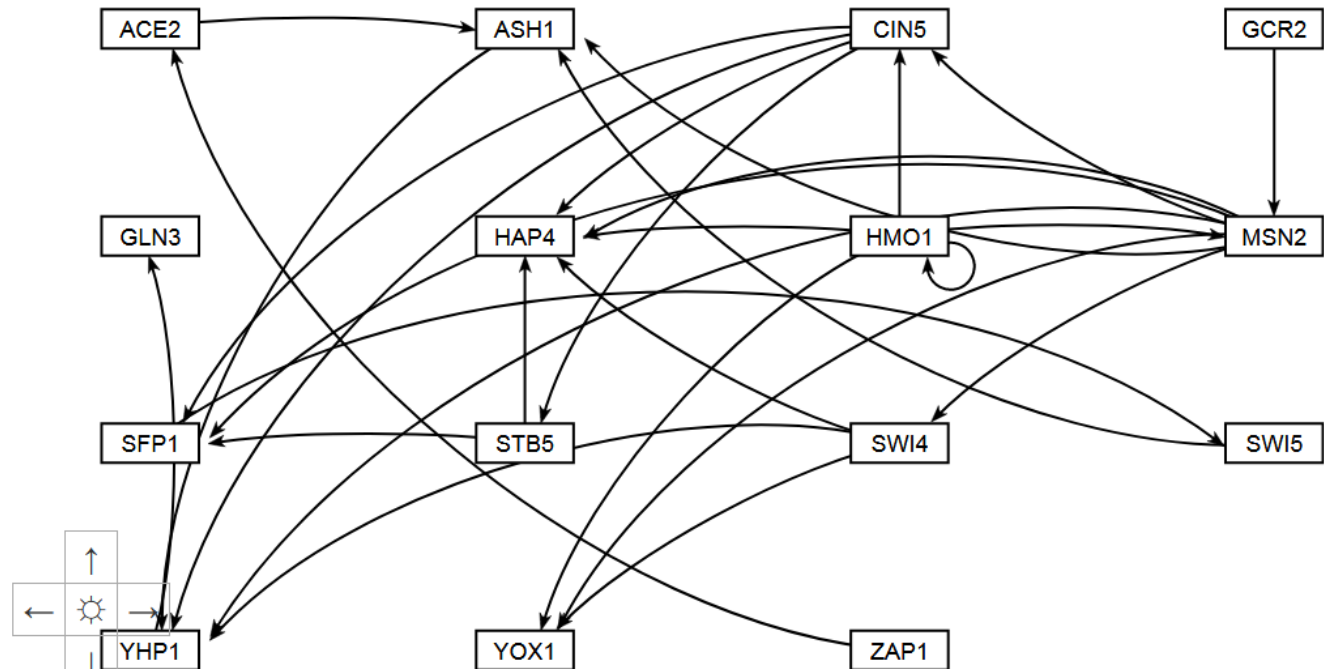
Reset Force Graph Parameters

Undo Reset

☒ Restrict graph to viewport

Viewport Size

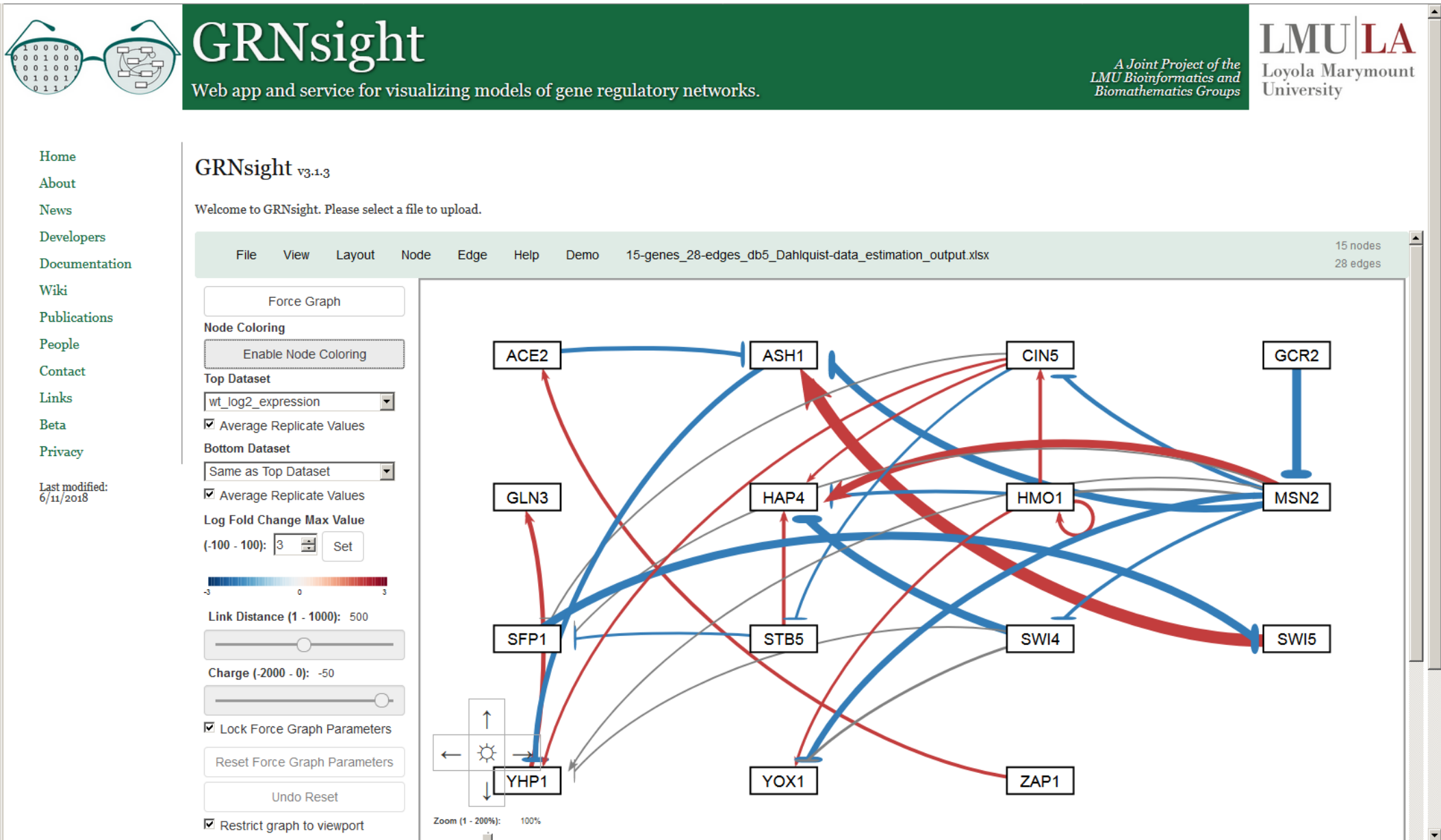
Small (1104 X 648 pixels)



Zoom (1 - 200%): 100%

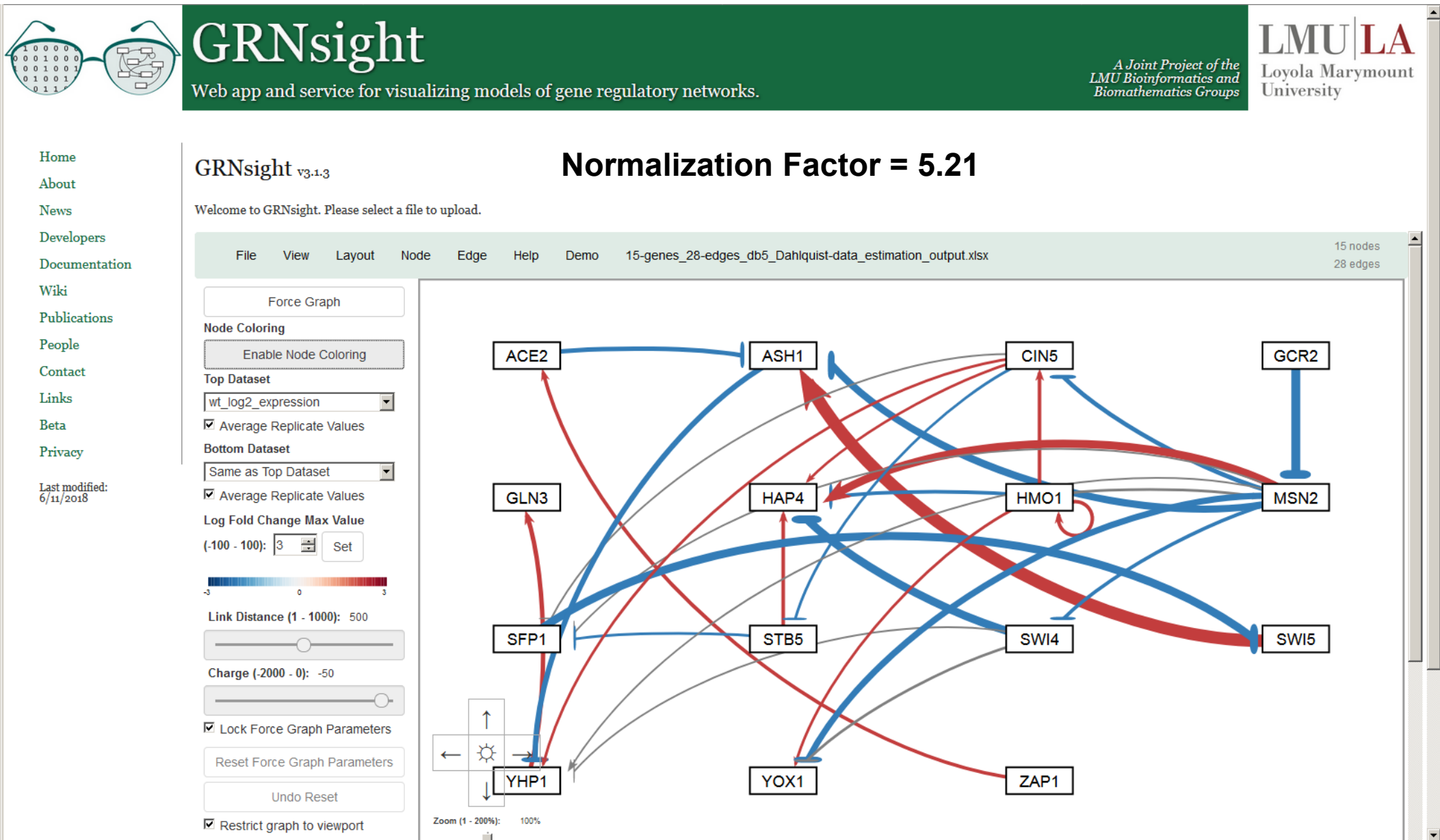
New in Version 3

When Model Results are Loaded, Edge Weight Values Are Indicated by Color, Thickness, and End Marker

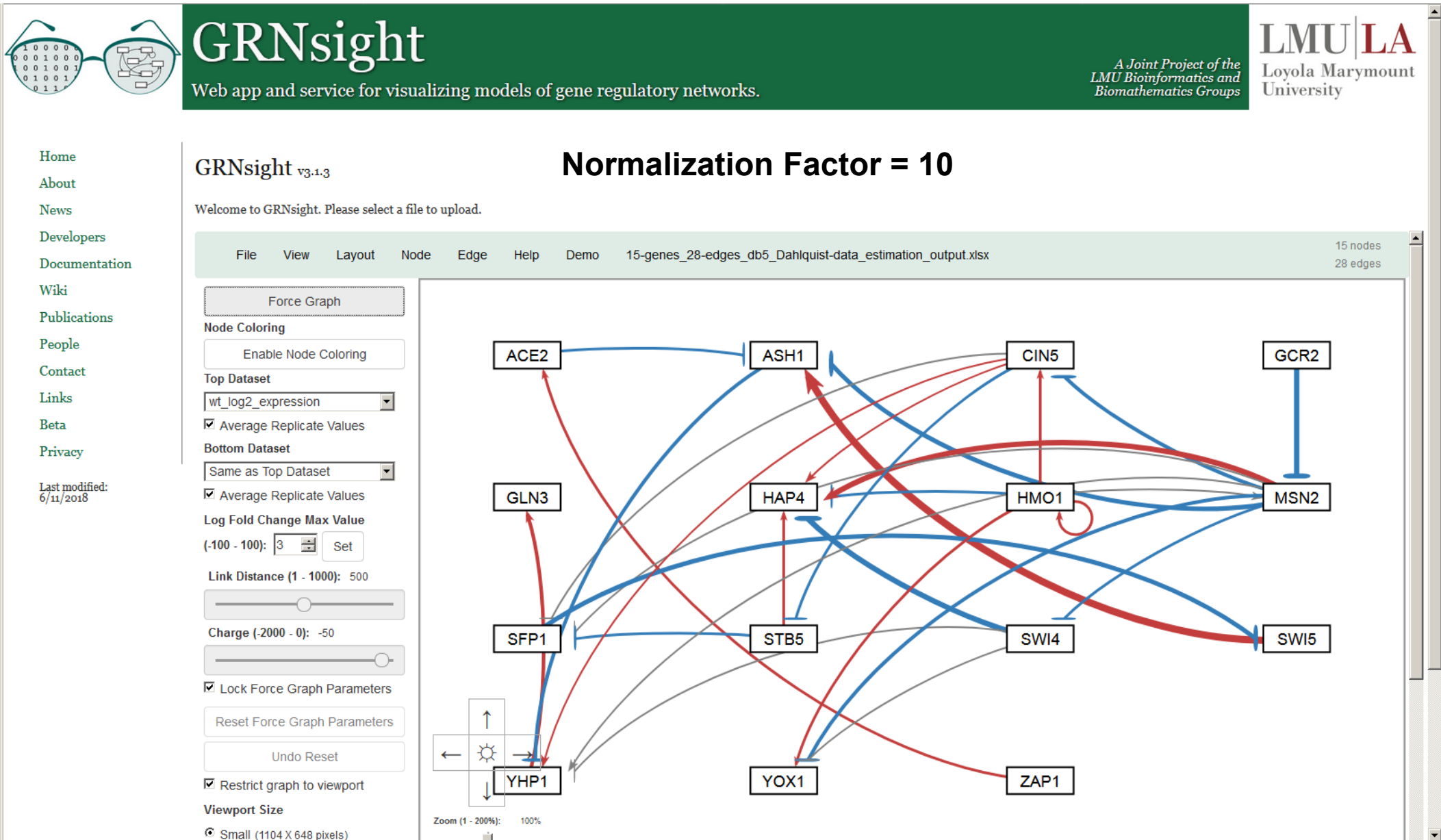


New color scheme in Version 3

Edge Thicknesses are Normalized to the Largest Magnitude Weight Value

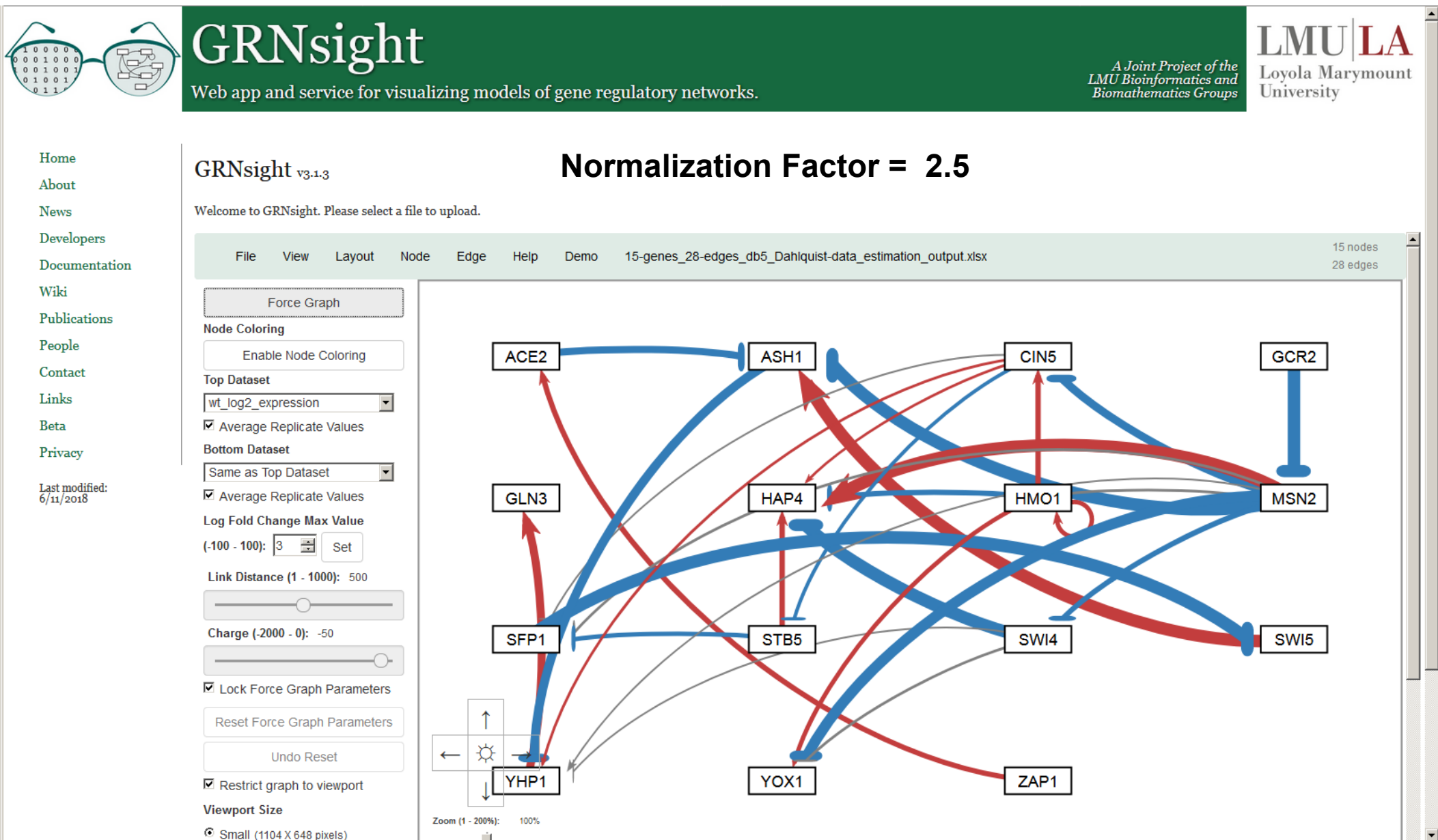


And Can Be Modified to Allow Fair Visual Comparison between Different Graphs



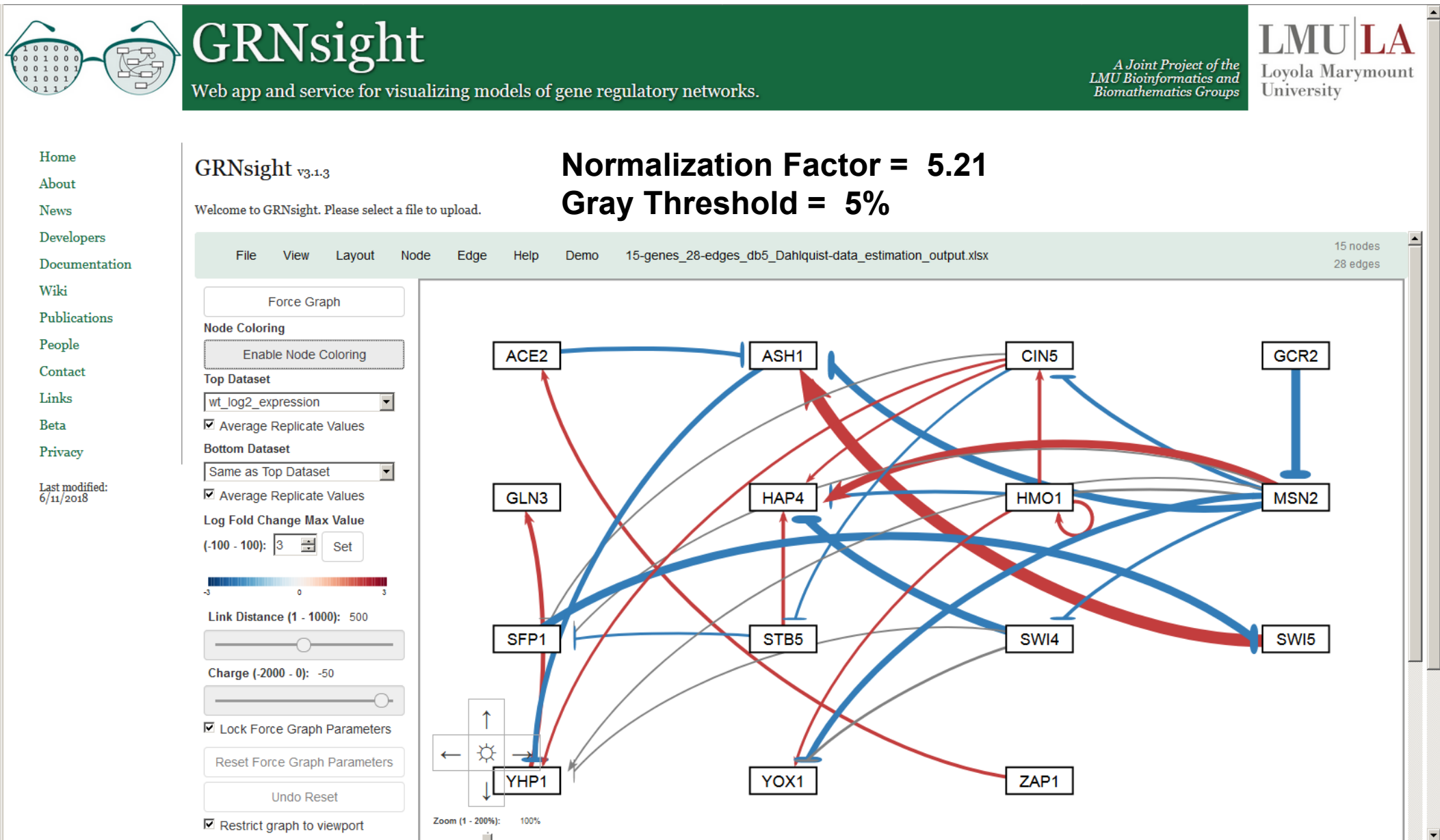
New in Version 2

And Can Be Modified to Allow Fair Visual Comparison between Different Graphs



New in Version 2

Edges with Normalized Weight Values within 5% of 0 Are Displayed as Gray to Minimize Their Visual Strength



Users Can Change the Gray Threshold with a Slider



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GRNsight v3.1.3

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Normalization Factor = 5.21

Gray Threshold = 10%

File View Layout Node Edge Help Demo 15-genes_28-edges_db5_Dahlquist-data_estimation_output.xlsx 15 nodes 28 edges

Force Graph

Node Coloring

Enable Node Coloring

Top Dataset

wt_log2_expression

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value

(-100 - 100): 3 Set

Link Distance (1 - 1000): 500

Charge (-2000 - 0): -50

☒ Lock Force Graph Parameters

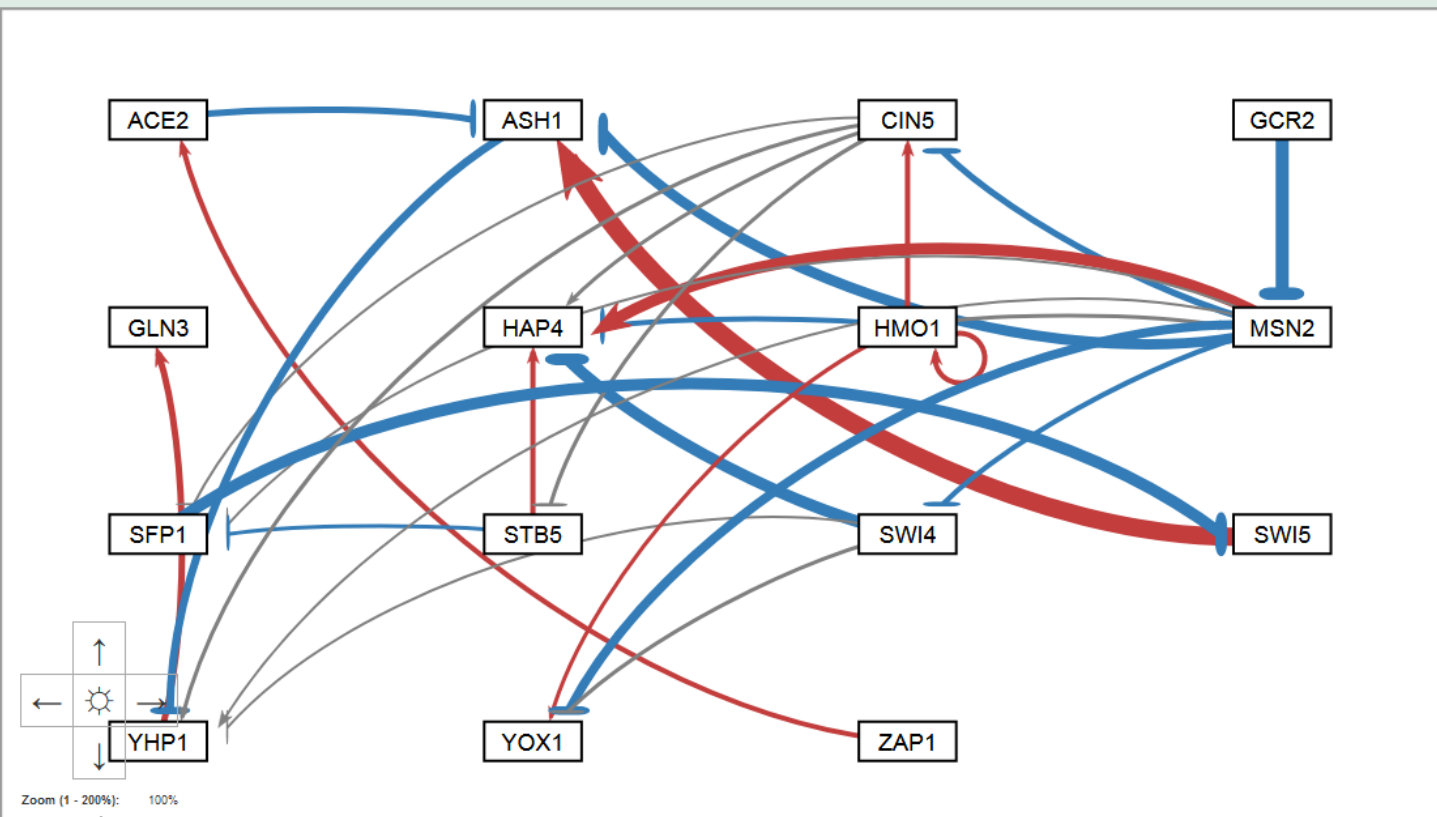
Reset Force Graph Parameters

Undo Reset

☒ Restrict graph to viewport

Viewport Size

Small (1104 X 648 pixels)



New in Version 2

Users Can Change the Gray Threshold with a Slider



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GRNsight v3.1.3

Welcome to GRNsight. Please select a file to upload.

Normalization Factor = 5.21
Gray Threshold = 25%

File View Layout Node Edge Help Demo 15-genes_28-edges_db5_Dahlquist-data_estimation_output.xlsx 15 nodes
28 edges

Force Graph

Node Coloring

Enable Node Coloring

Top Dataset

wt_log2_expression

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value

(-100 - 100): 3 Set

Link Distance (1 - 1000): 500

Charge (-2000 - 0): -50

☒ Lock Force Graph Parameters

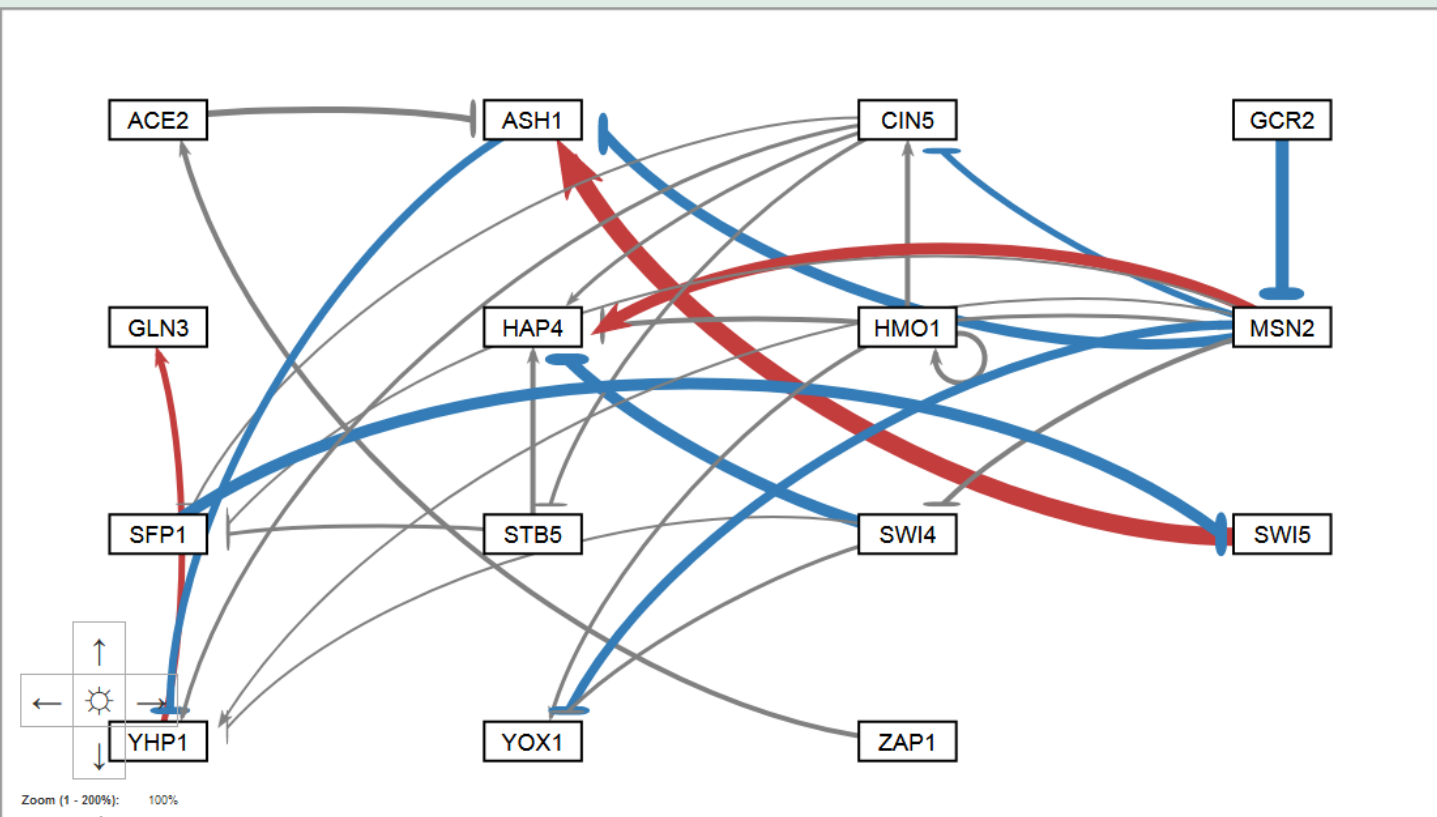
Reset Force Graph Parameters

Undo Reset

☒ Restrict graph to viewport

Viewport Size

Small (1104 X 648 pixels)



Zoom (1 - 200%): 100%

New in Version 2

Users Can Change the Gray Threshold with a Slider



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GRNsight v3.1.3

Welcome to GRNsight. Please select a file to upload.

Normalization Factor = 5.21

Gray Threshold = 50%

FileViewLayoutNodeEdgeHelpDemo15-genes_28-edges_db5_Dahlquist-data_estimation_output.xlsx15 nodes28 edges

Force Graph

Node Coloring

Enable Node Coloring

Top Dataset

wt_log2_expression

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value

(-100 - 100): 3Set

Link Distance (1 - 1000): 500

Charge (-2000 - 0): -50

☒ Lock Force Graph Parameters

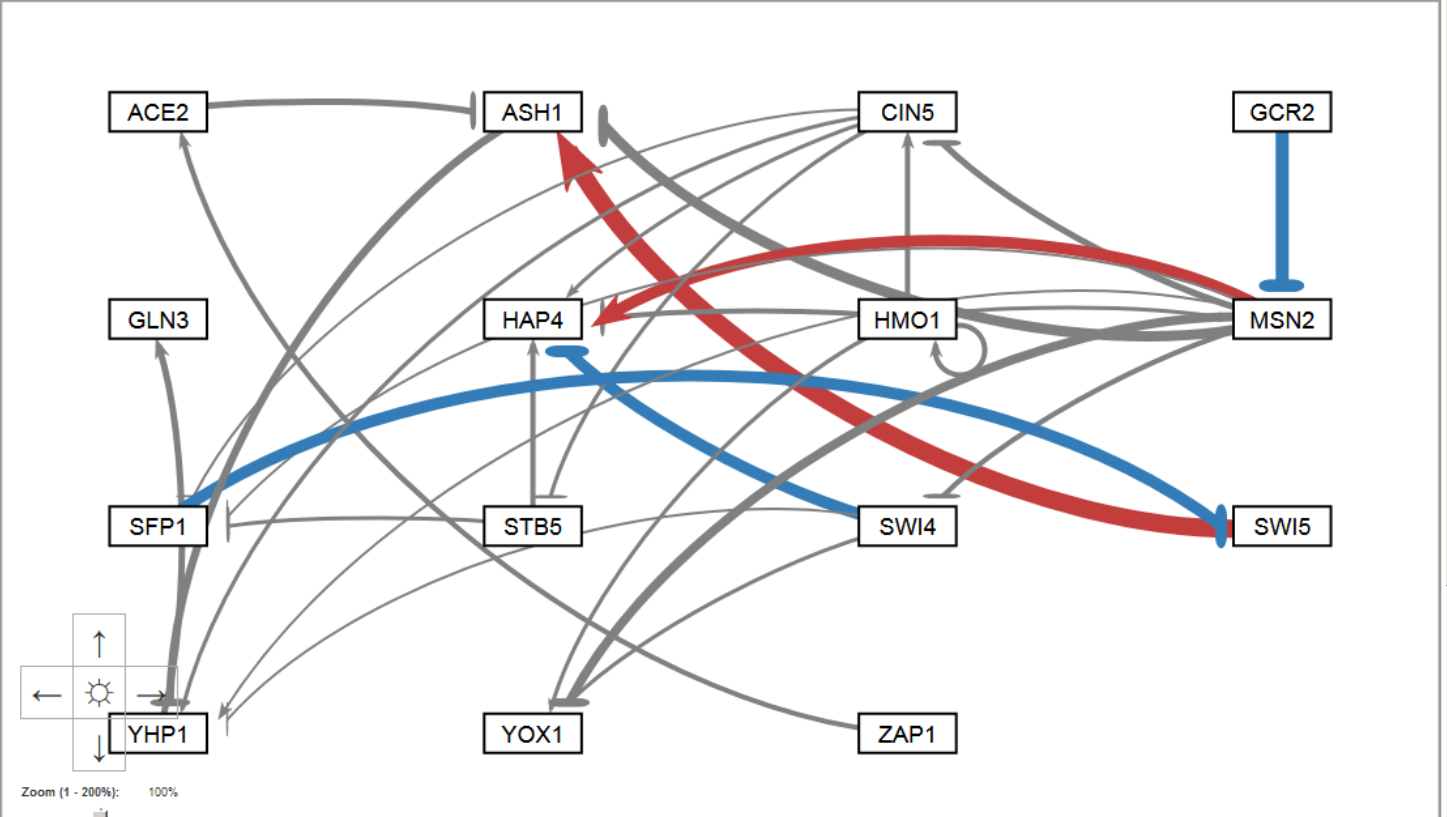
Reset Force Graph Parameters

Undo Reset

☒ Restrict graph to viewport

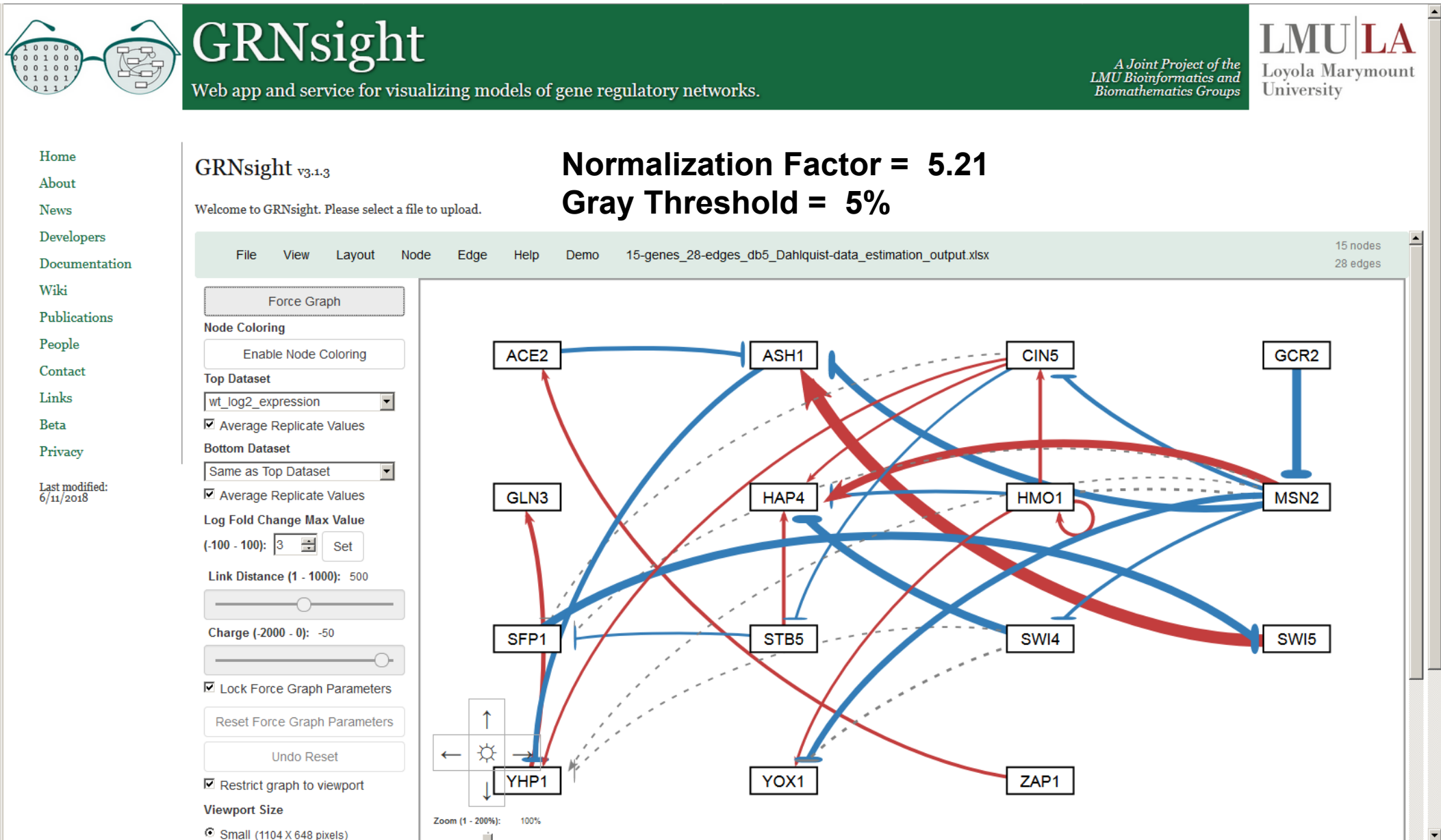
Viewport Size

Small (1104 X 648 pixels)



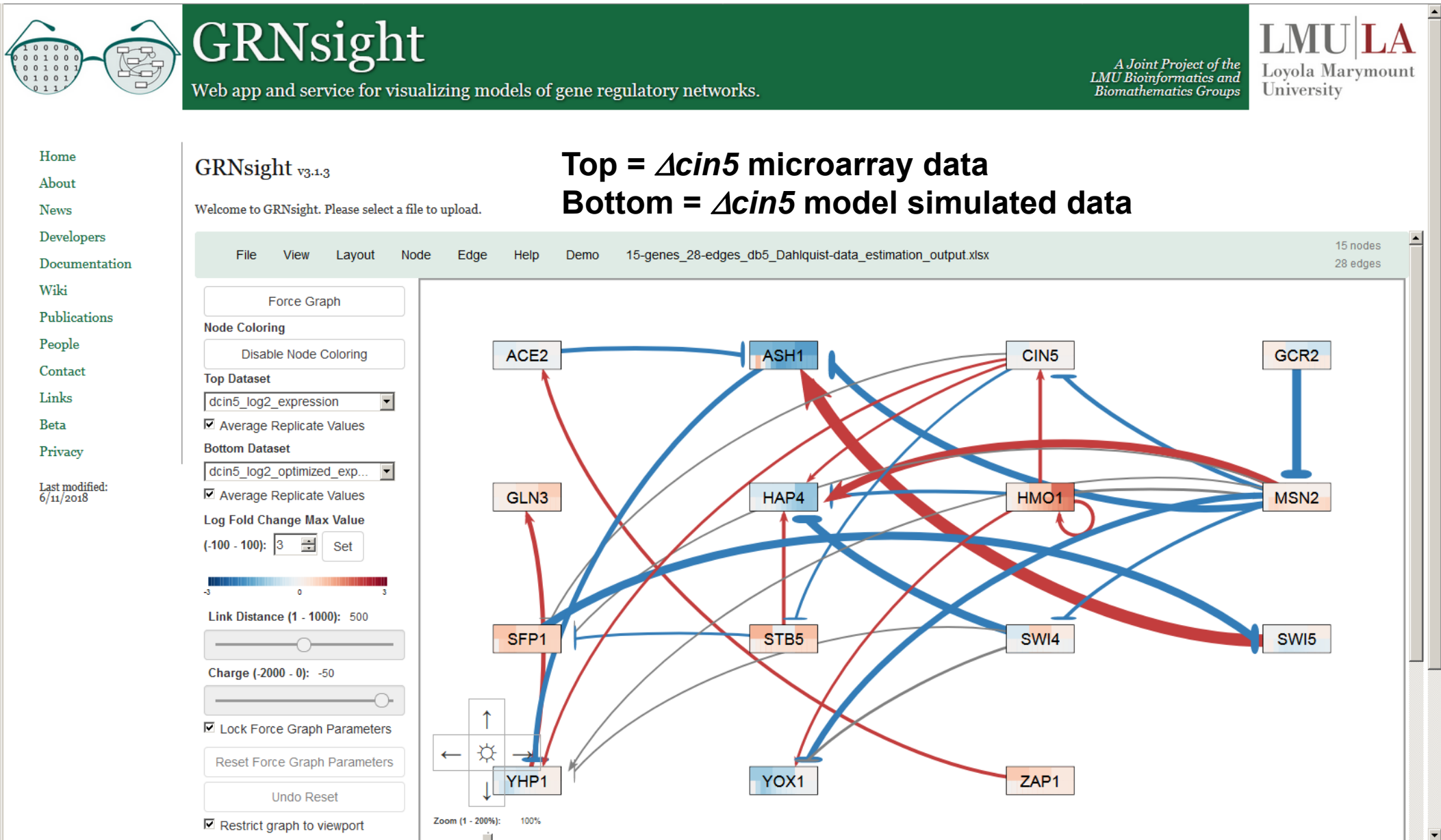
New in Version 2

Gray Edges Can Also Be Shown as Dashed Lines to Facilitate Visualization by Color Blind Users

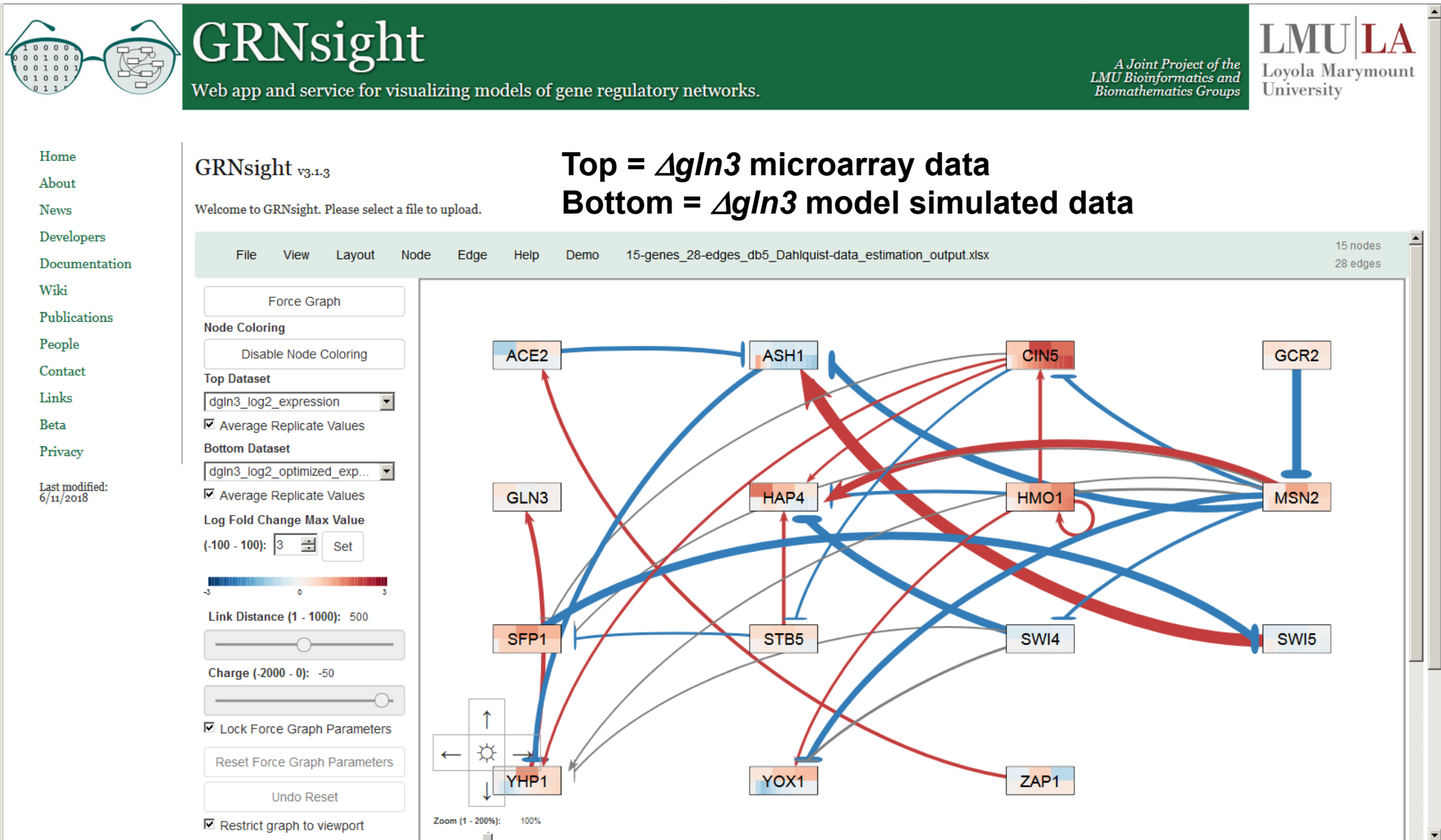


New in Version 3

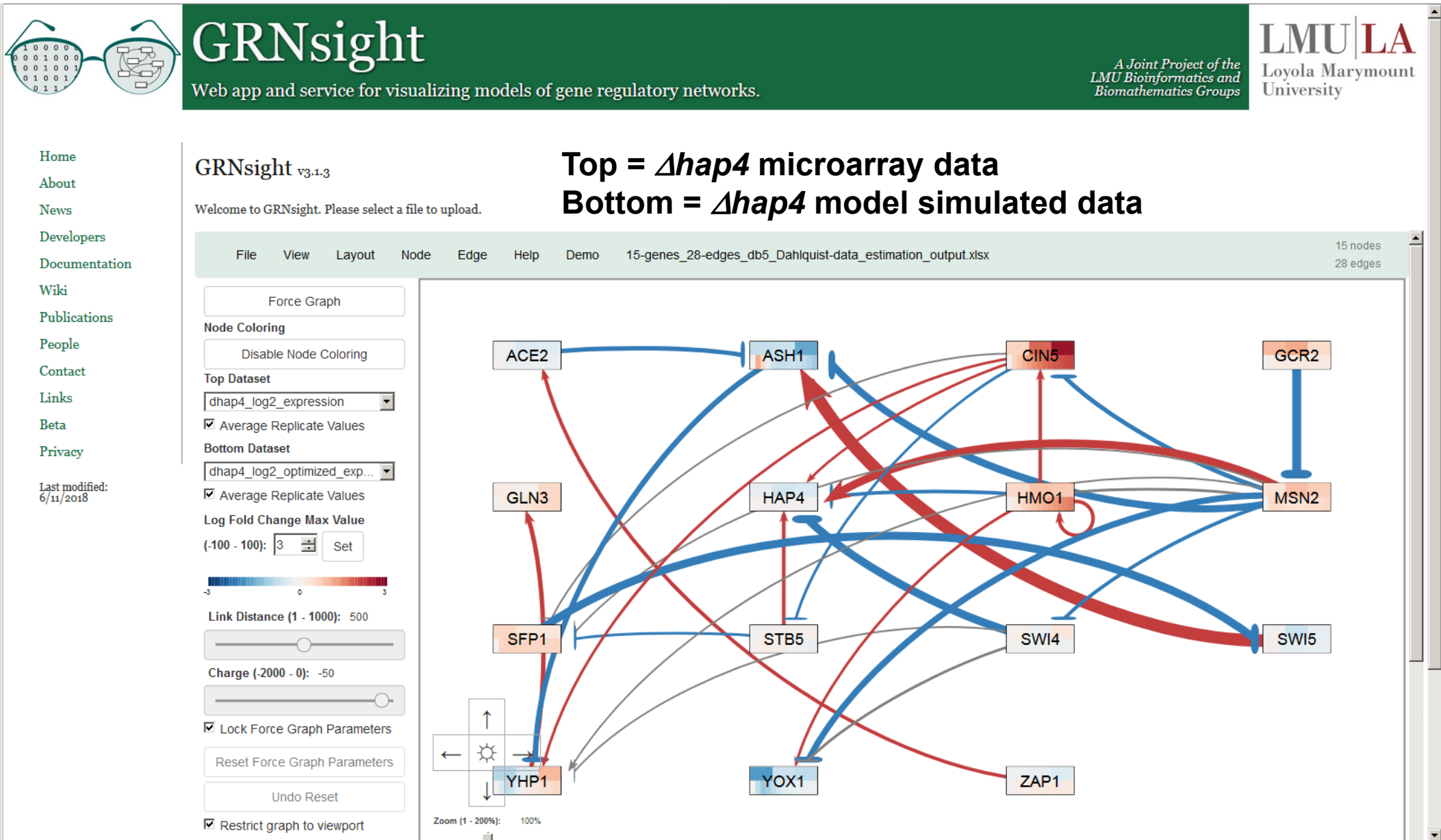
Generally, We Can See a Good Concordance Between the Measured and Simulated Data for Each Strain



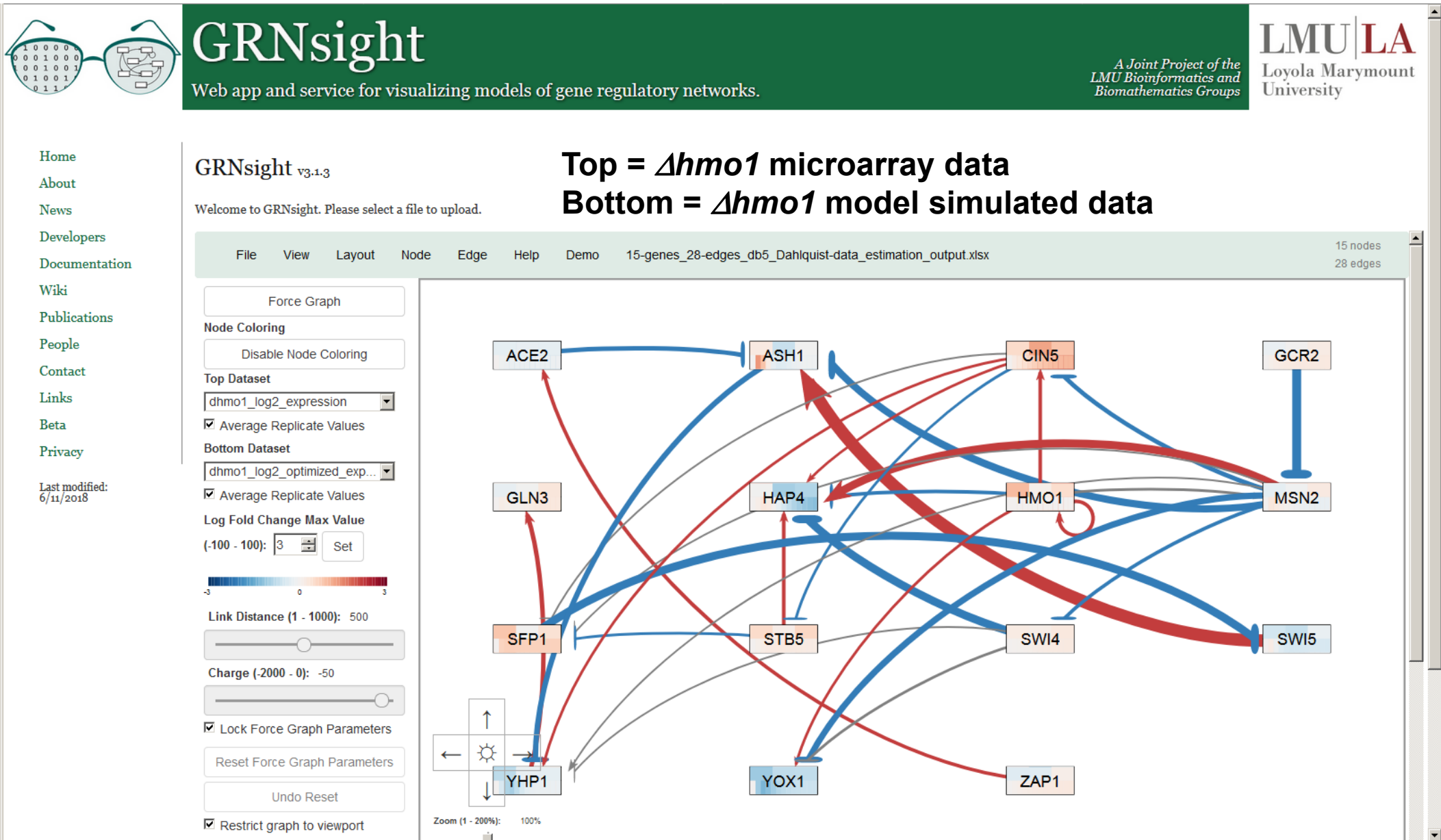
Generally, We Can See a Good Concordance Between the Measured and Simulated Data for Each Strain



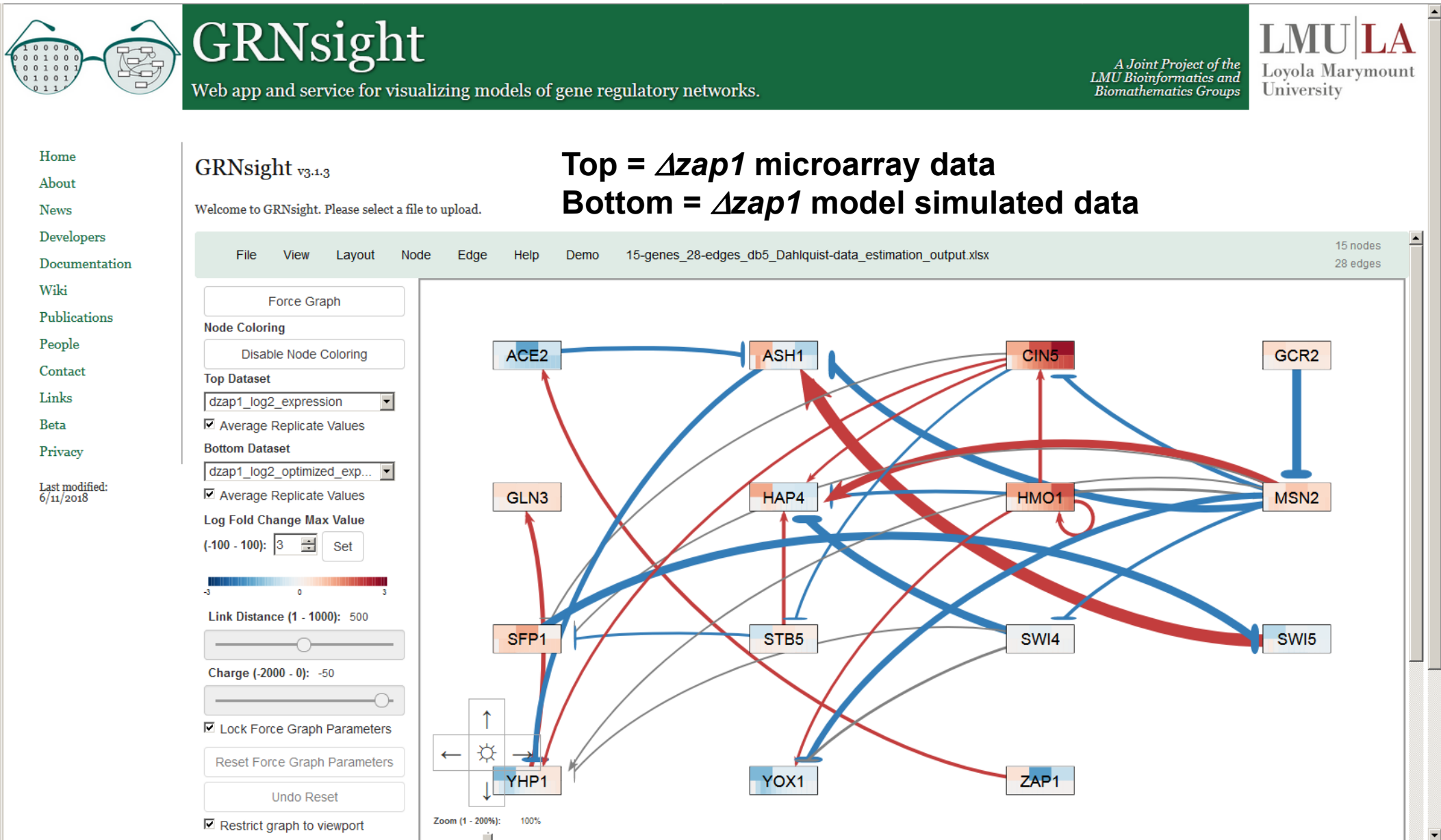
Generally, We Can See a Good Concordance Between the Measured and Simulated Data for Each Strain



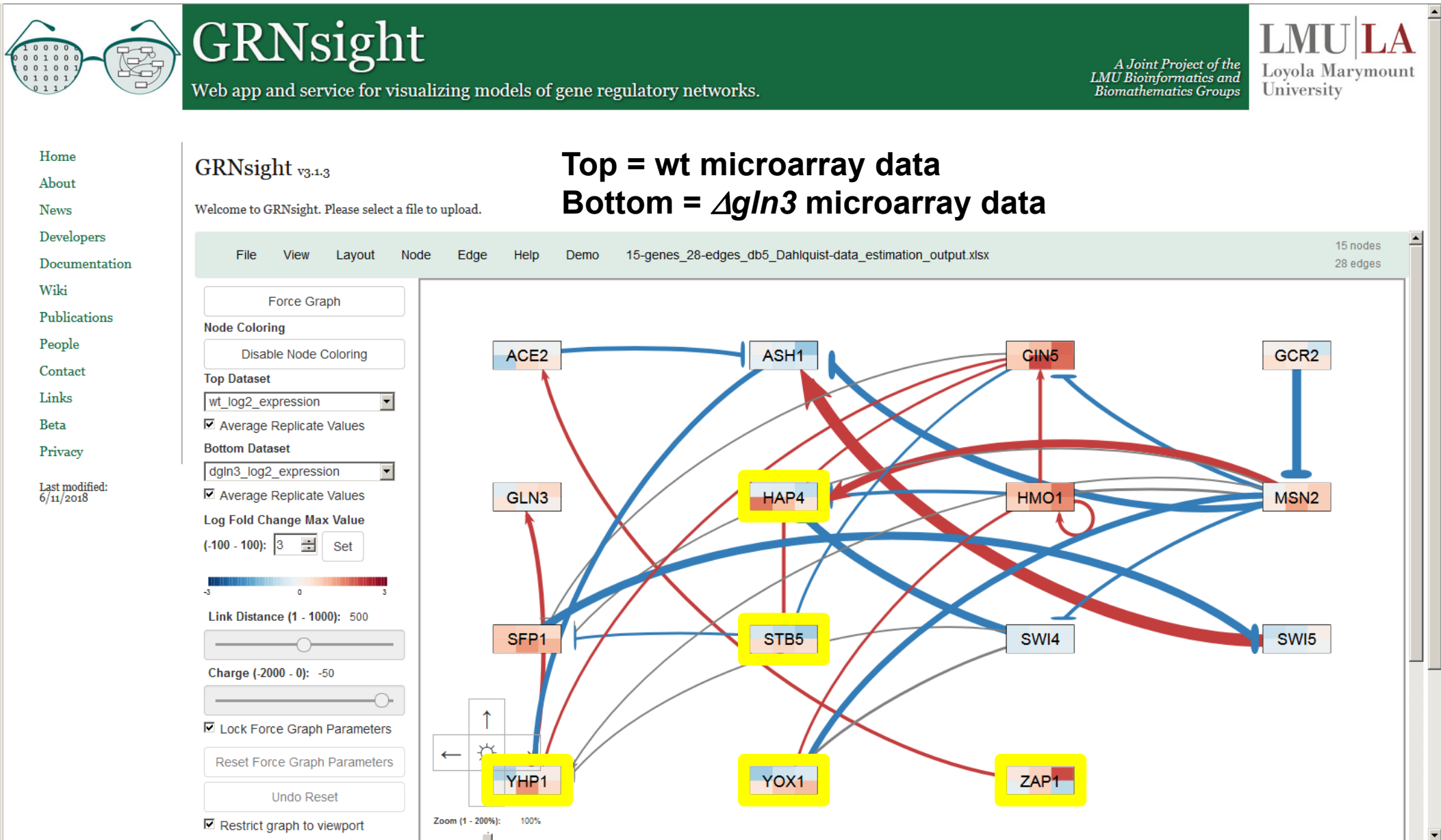
Generally, We Can See a Good Concordance Between the Measured and Simulated Data for Each Strain



Generally, We Can See a Good Concordance Between the Measured and Simulated Data for Each Strain



We Can Also See that There Are Notable Differences in Expression Between Strains



Right-clicking on a Gene Opens a New Web Page with Information Dynamically Retrieved from Source Databases



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Last modified:
5/18/2018

GLN3

Saccharomyces cerevisiae (strain ATCC
204508 / S288c)

General Information

Protein Information

Regulation

Gene Ontology

Sources

General Information

SGD ID:	S000000842 ^[1]
NCBI Gene ID:	856763 ^[4]
Ensembl ID:	Not found ^[3]
Uniprot ID:	GLN3_YEAST ^[2]
JASPAR ID:	MA0307.1 ^[5]
Description:	Transcriptional activator of genes regulated by nitrogen catabolite repression; localization and activity regulated by quality of nitrogen source and Ure2p ^[1]
Species:	<i>Saccharomyces cerevisiae</i> (strain ATCC 204508 / S288c) ^[2]
Locus Tag:	YER040W ^[4]
JASPAR Family:	GATA-type zinc fingers ^[5]
JASPAR Class:	Other C4 zinc finger-type factors ^[5]
Chromosome Sequence:	NC_001137.3 ^[4]

Protein Information

Regulation

Gene Ontology

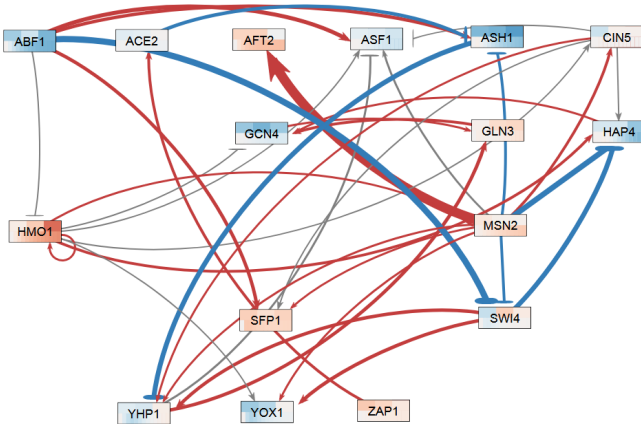
Sources

New in Version 3

Back to DB1-DB6: How Well Do Their Models Perform Relative to Each Other?

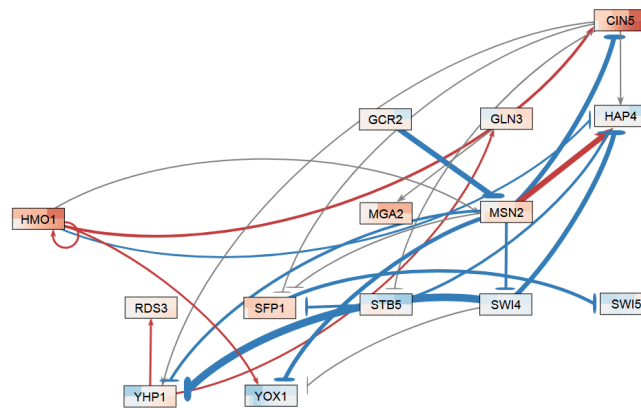
db1: 16 Genes, 26 Edges

LSE:minLSE = 1.42



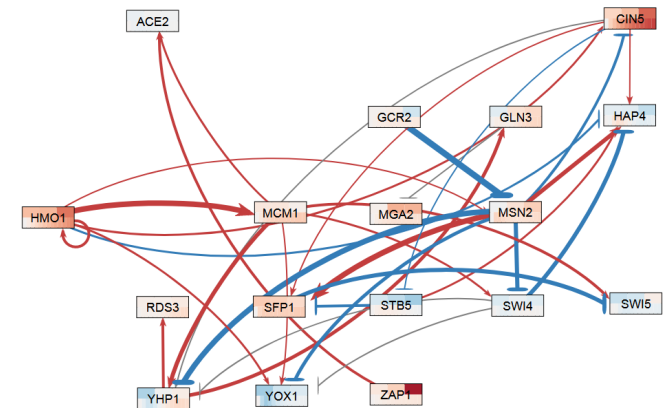
db2: 14 Genes, 25 Edges

LSE:minLSE = 1.36



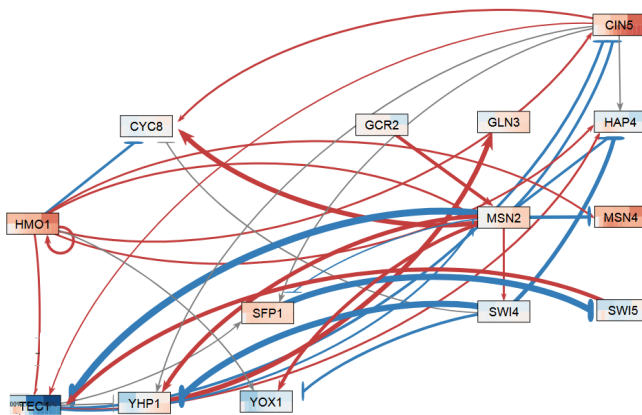
db3: 17 Genes, 32 Edges

LSE:minLSE = 1.41



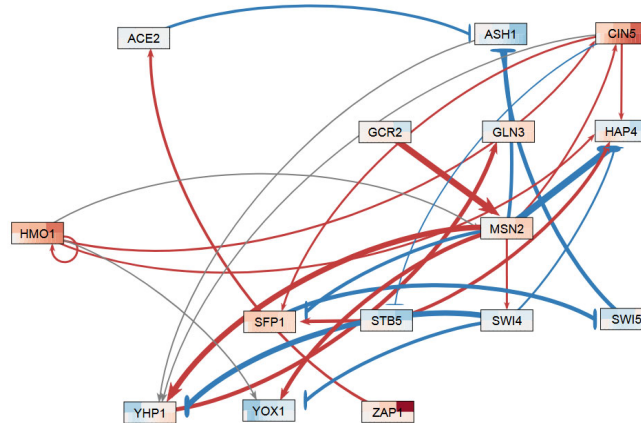
db4: 14 Genes, 35 Edges

LSE:minLSE = 1.30



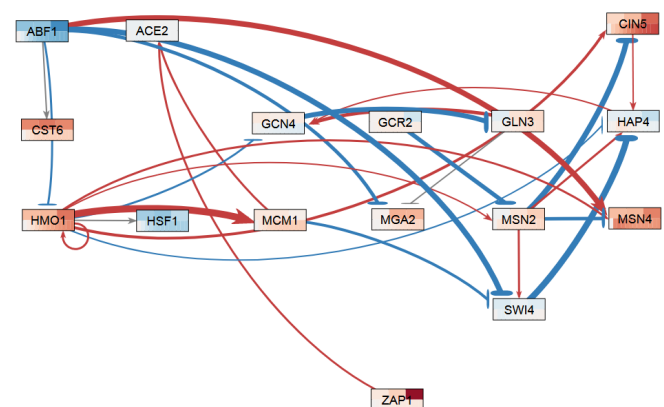
db5: 15 Genes, 28 Edges

LSE:minLSE = 1.43



db6: 16 Genes, 27 Edges

LSE:minLSE = 1.40



Multiple Regression Analyses Identified Determinants of Individual Transcription Factor Model Fits

	Database-Derived Networks						3 Best Random Networks			3 Worst Random Networks		
	db1	db2	db3	db4	db5	db6	rand15	rand16	rand24	rand7	rand12	rand31
ANOVA B&H p-value												
Average Expression at 15 min.	-0.507	-0.624	-0.479	-0.552	-0.528	-0.626				-0.838		-0.656
Average Expression at 30 min.												
Average Expression at 60 min.											-0.503	
Degradation Rate							-0.505		-0.611			
Optimized Production Rate	0.578			0.777		0.437						
Optimized Threshold b				0.883								
Weighted In-degree						0.617						
Weighted Out-degree							-0.627			0.433		
Eccentricity												
Closeness Centrality							-0.443					
Betweenness Centrality								-0.609		0.399		
Clustering		0.522	0.473		0.501							
Eigenvector Centrality	-0.384			-0.533		-0.328			-0.598		-0.685	-0.831
Modularity Class	—	—	—	—	—	—					0.510	
Page Ranks	—	—	—	—	—	—						0.633
Strong Component Number	—	—	—	—	—	—		0.633			-1.033	
Adjusted R ²	0.290	0.525	0.360	0.441	0.350	0.196	0.436	0.552	0.257	0.500	0.692	0.437

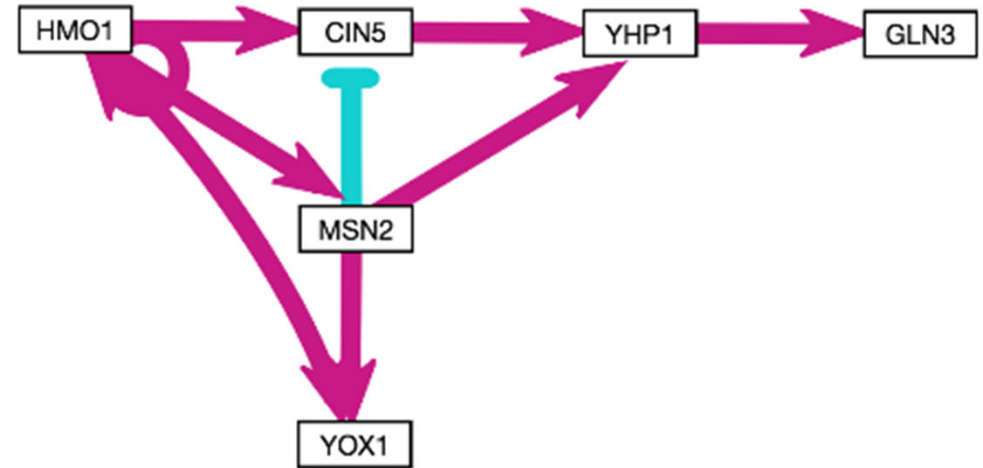
- Factors that were **not significant** predictors of MSE:minMSE ratio:
ANOVA B&H p-value, Average Expression at 30 minutes, Eccentricity
- Factors that were **consistently significant** predictors of MSE:minMSE ratio:
Average Expression at 15 minutes, Eigenvector Centrality

Motifs Conserved in the Database-Derived Networks Were Identified

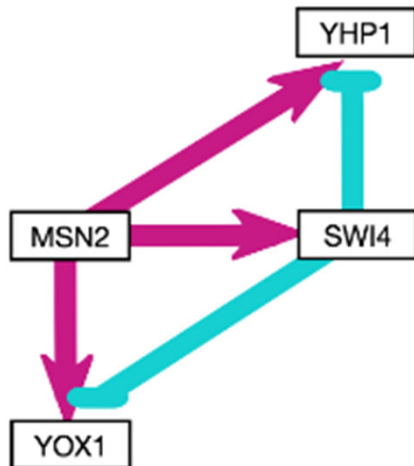
A



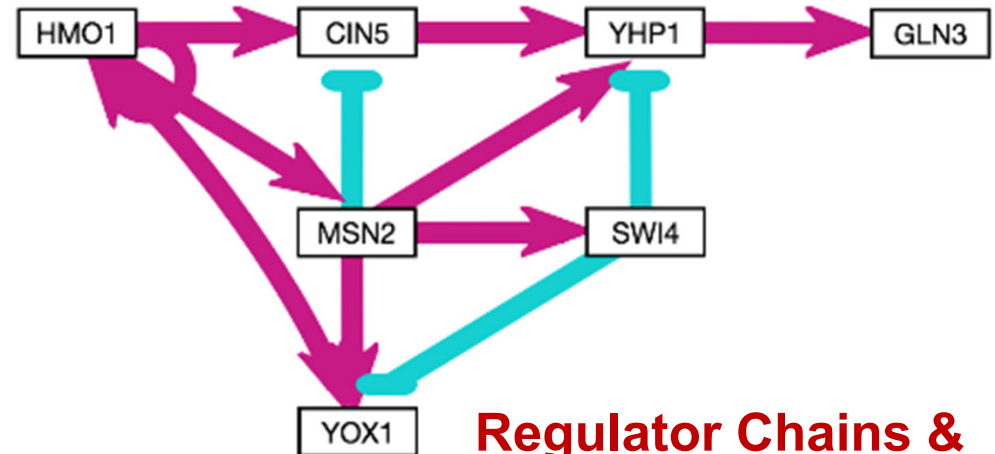
B



C



D



**Regulator Chains &
5 Feedforward Loops**